University of Amsterdam
Amsterdam Business School

Machine Learning with Python

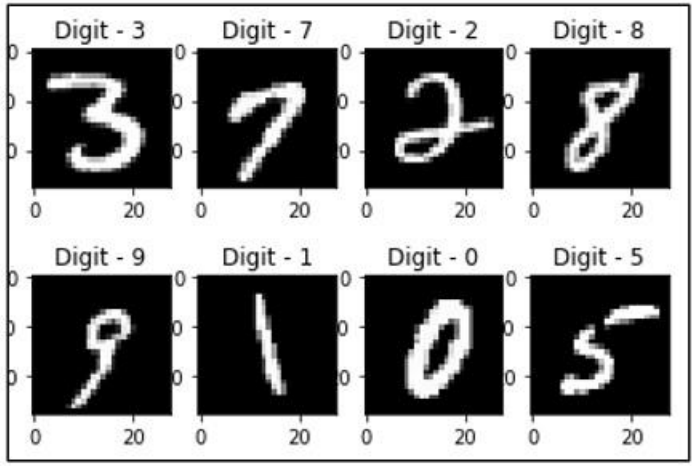
AAA Global Connect 2026: DATA Workshop

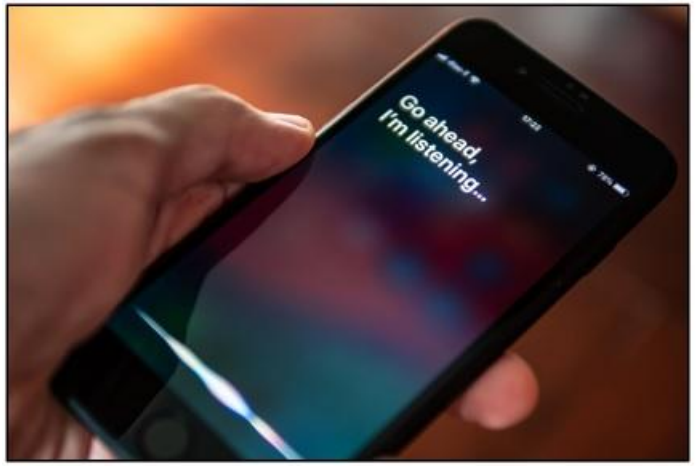
Dr. Jeroen van Raak

Outline

- Machine learning applications
- Definition of machine learning
- Types of machine learning
- Illustrations:
  - k-Nearest Neighbors
  - Decision Tree
  - Artificial Neural Network
  - k-Means

Machine Learning Applications





UvA

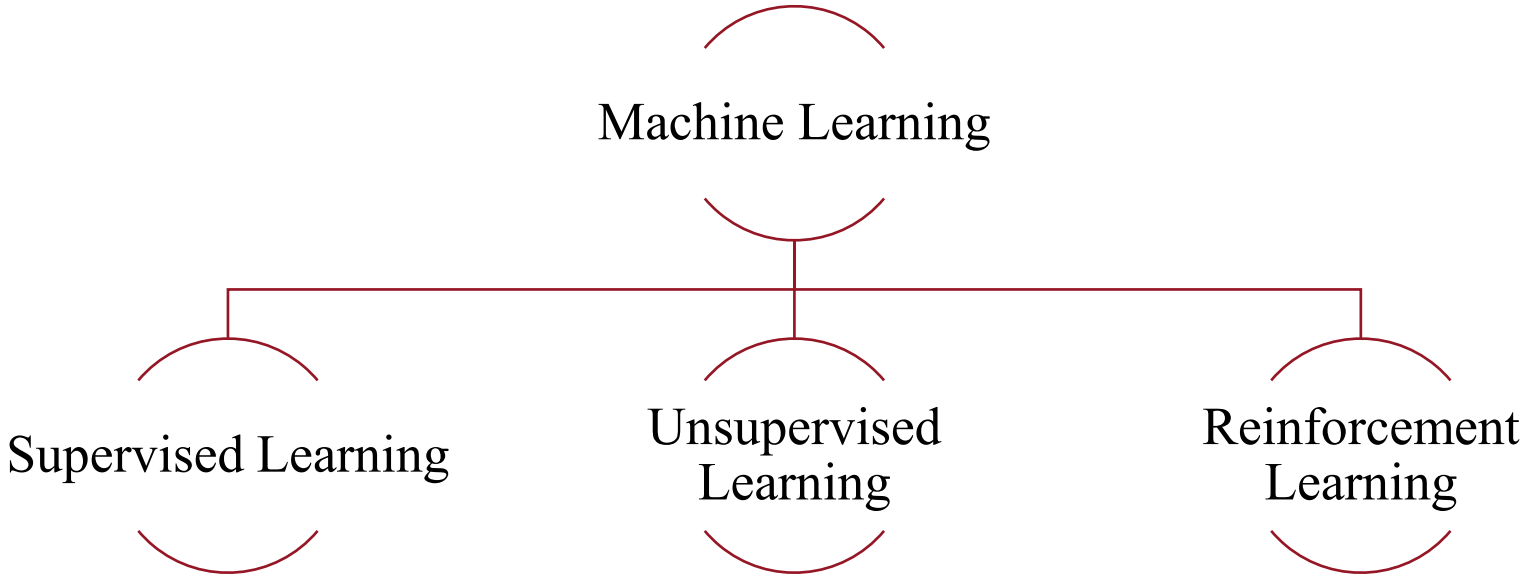
Other Machine Learning Applications

- Spam filtering
- Summarizing documents
- Creating chatbots
- Forecasting sales
- Fraud and anomaly detection
- Segmenting clients
- Product recommendations
- Moderating forum discussions
- Bots for games
- Generating code, text or images

Machine Learning Definitions

- Machine learning: field of study that gives computers the ability to learn without being explicitly programmed (Arthur Samuel, 1959)
- Machine learning is the technique that improves system performance by learning from experience via computational methods. In computer systems, experience exists in the form of data, and the main task of machine learning is to develop learning algorithms that build models from data. (Zhou, 2021)

Machine Learning Paradigms



- Uses **labeled data** (input-output pairs) to train the model.
- Learns to predict outcomes for new, unseen inputs.
- Commonly used for classification (e.g., fraud detection) and regression (e.g., risk scoring).

- Works with **unlabeled data** to discover patterns or groupings.
- Identifies clusters, anomalies, or structures in data.
- Useful for anomaly detection or client segmentation in auditing.

- Involves an agent that **learns through trial and error**, receiving rewards or penalties.
- Learns a strategy (policy) to maximize long-term rewards.
- Used in decision-making and optimization tasks.

UvA

Machine Learning Algorithms

- Supervised:
  - *k*-Nearest Neighbors
  - Linear regression
  - Logistic regression
  - Support Vector Machines
  - Naive Bayes
  - Decision Trees and Random Forests
  - Neural networks
- Unsupervised:
  - Clustering (e.g., *k*-means or DBSCAN)
  - Reinforcement learning
  - Principal Component Analysis
    (i.e., reduce the number of variables)

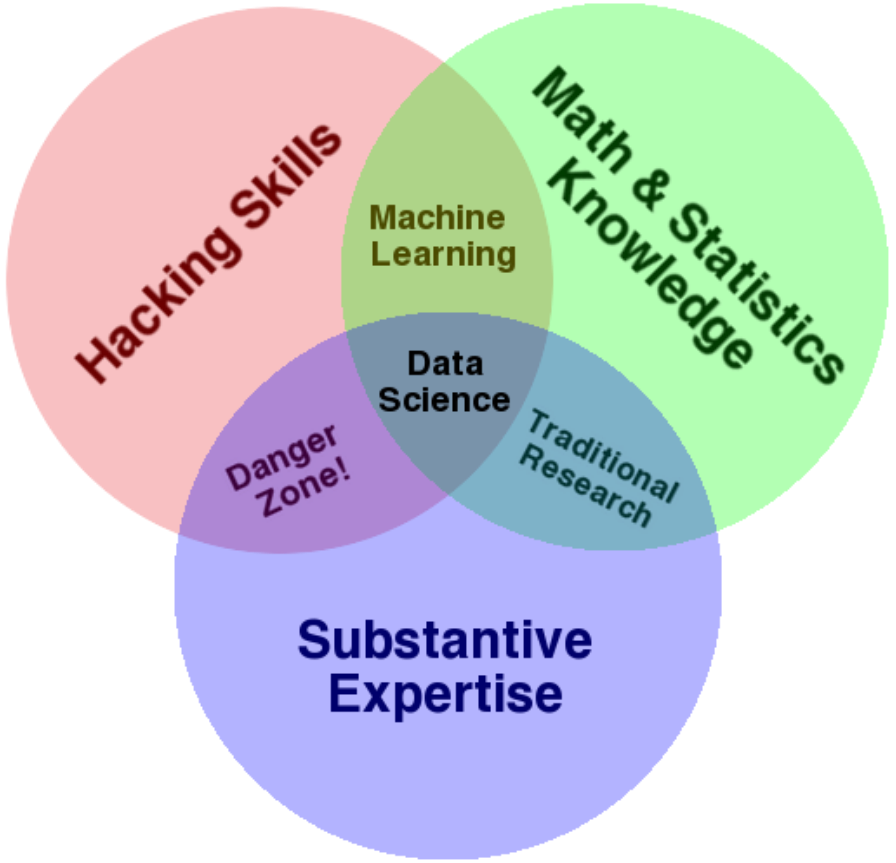
(Gong 2022)

UvA

Machine Learning Strategies

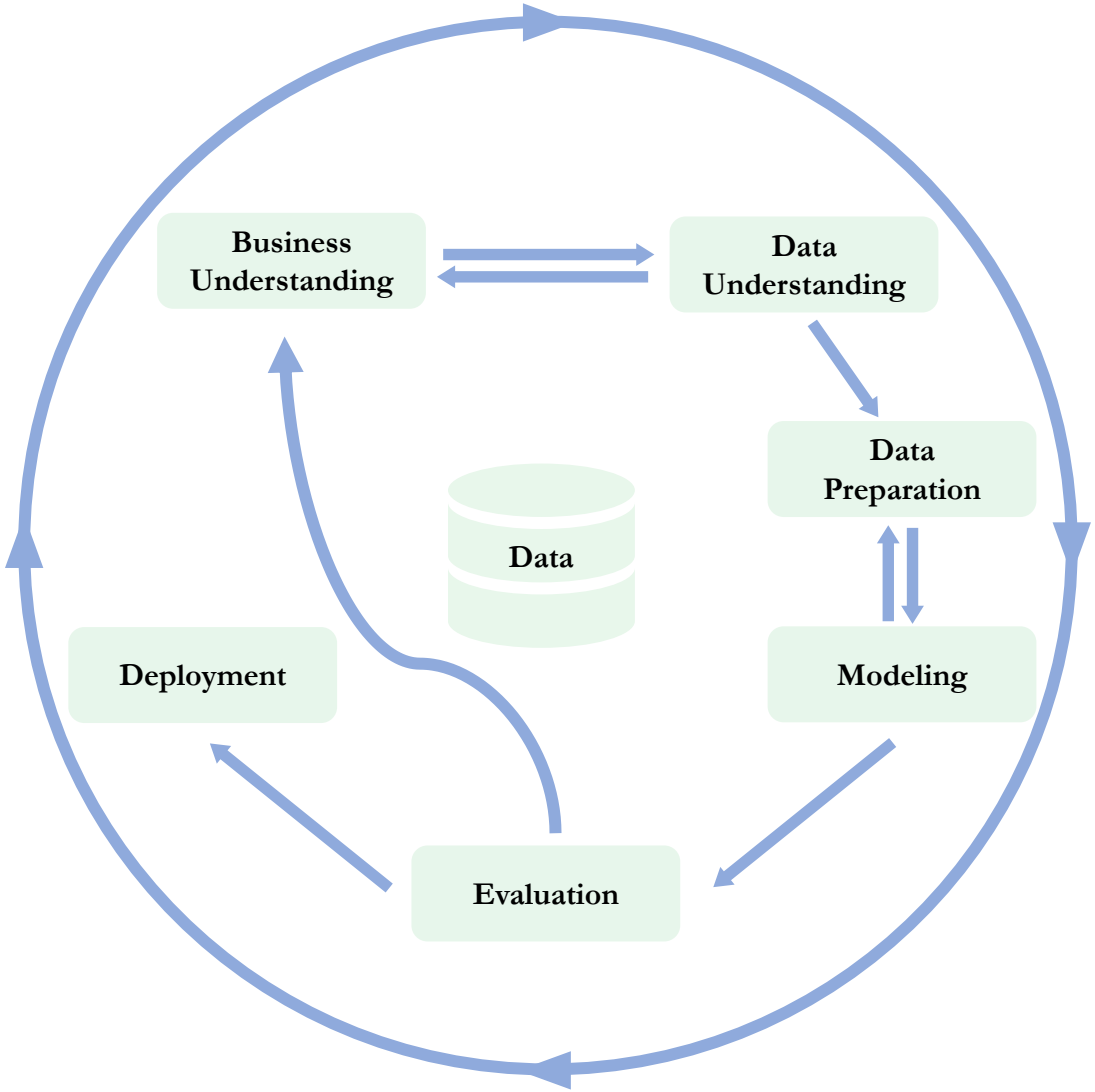
- Online vs. Batch Learning
  - Batch Learning: trains on entire dataset, heavy computation, retrain for new data
  - Online Learning: incremental updates, adapts quickly
- Instance-Based vs. Model-Based Learning
  - Instance-Based: memorizes examples, compares new data
  - Model-Based: learns general model, fast predictions, scalable

Data Science Venn Diagram



Source: http://drewconway.com/zia/2013/3/26/the-data-science-venn-diagram

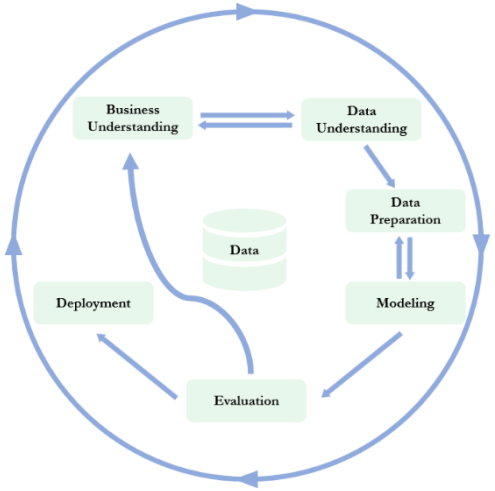
Machine Learning Framework: CRISP-DM



Adapted from Chapman et al. (2000)

UvA

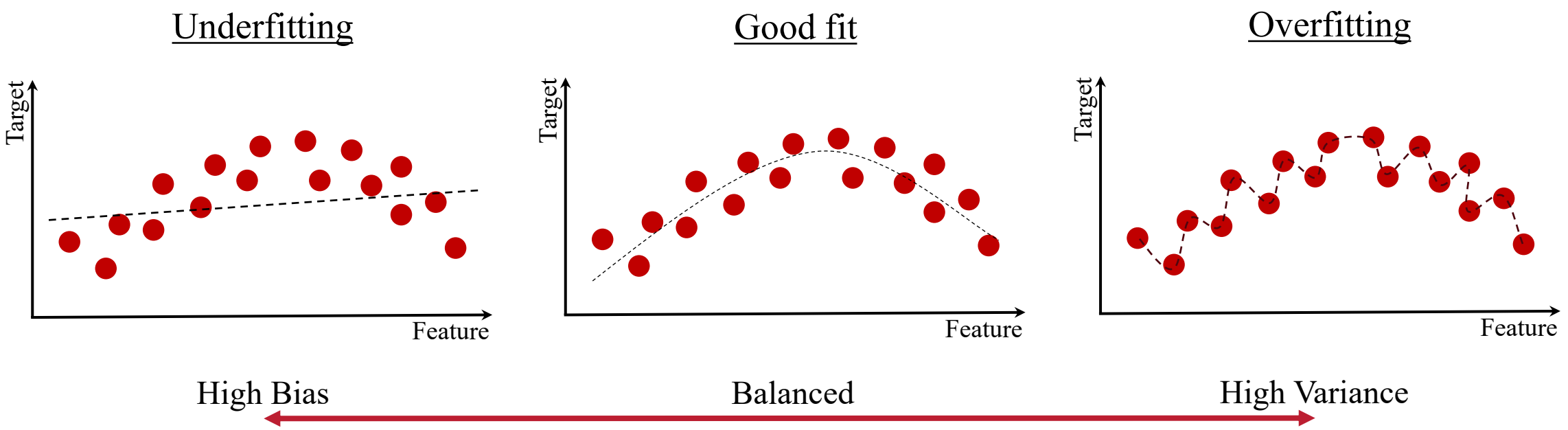
Data Preparation vs Analysis Time



- Important distinction between tasks:
  1. Data preparation: cleaning, EDA, feature engineering
  2. Analysis: algorithm selection, model training, evaluation
- Distribution of time allocated to these tasks for a typical project:

- Preparing data is often the most time-consuming and iterative phase
  - Allocate sufficient time, it directly impacts model quality

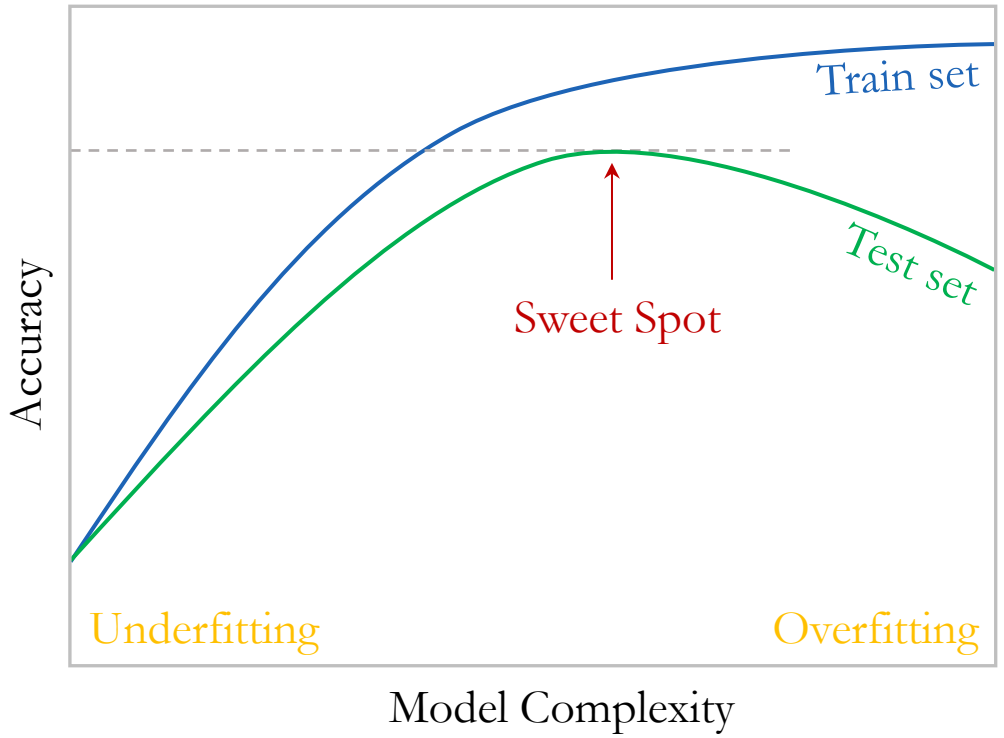
Model Complexity: Bias vs Variance



Bias measures how far off, on average, a model's predictions are from the true values.
Variance measures how much a model's predictions change with different training datasets.

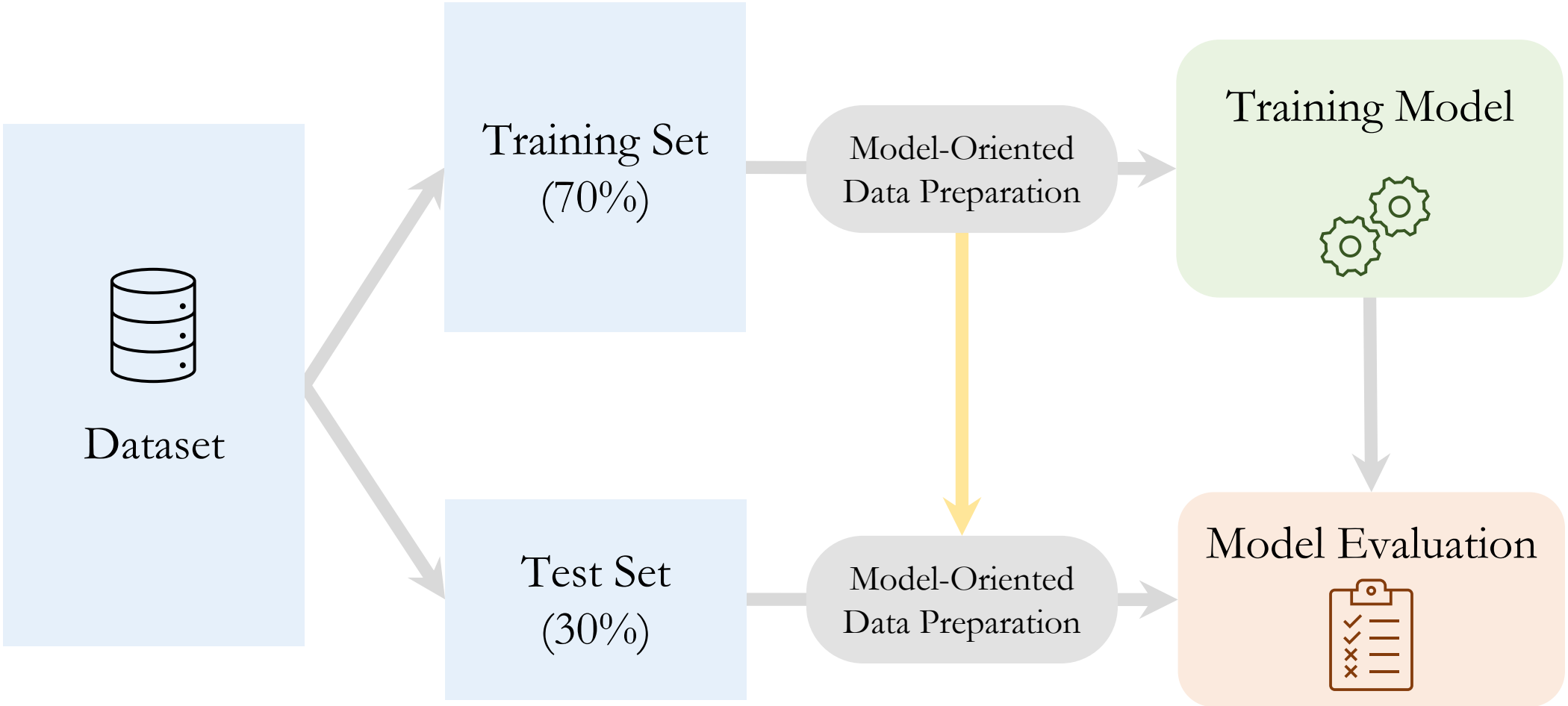
UvA

Model Complexity: Overfitting vs Underfitting

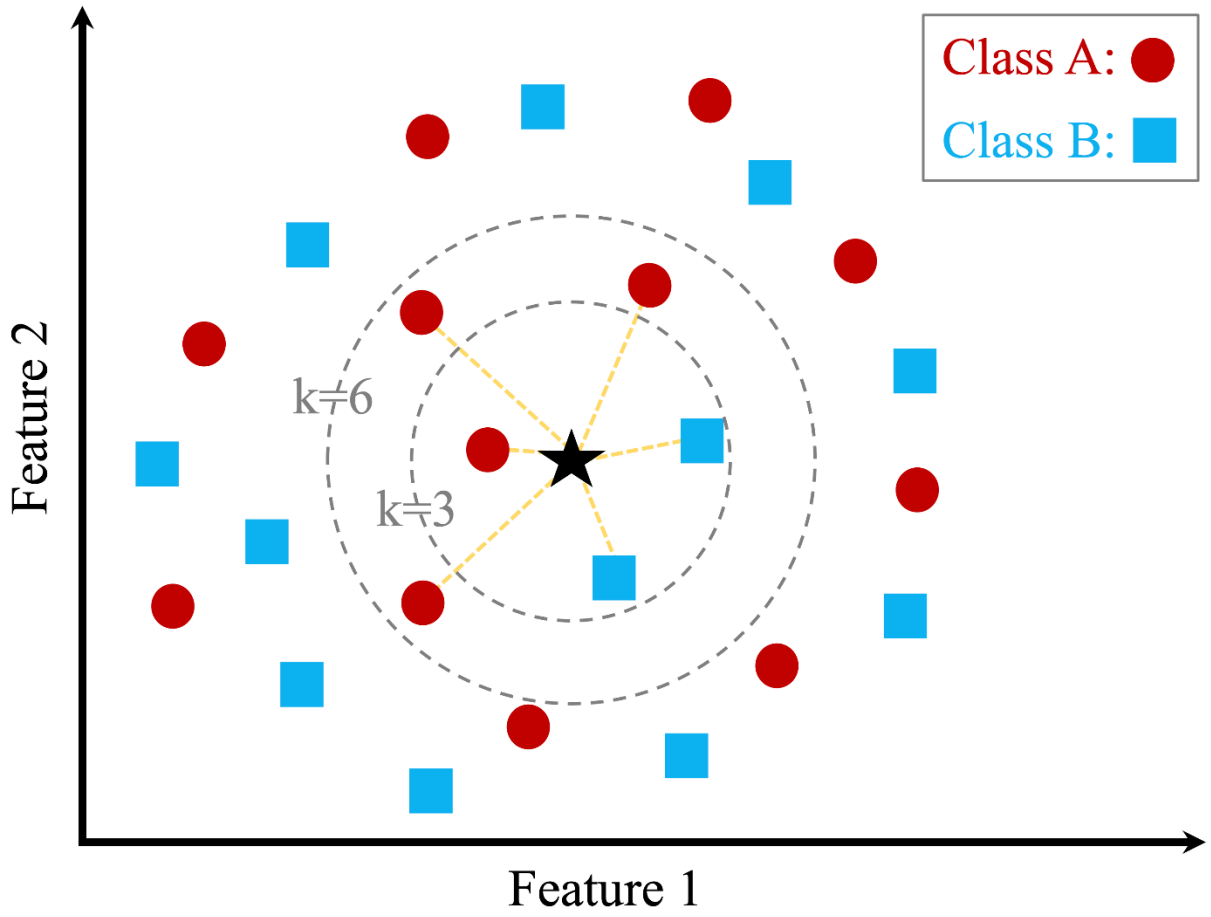


UvA

Data Preparation and Modeling



k-Nearest Neighbors Algorithm



For k = 3:
Selected class is **B** with 2/3=66% confidence

For k = 6:
Selected class is **A** with 4/6=66% confidence

k-Nearest Neighbor Algorithm

The *k*-Nearest Neighbor algorithm is an instance-based learning where training set records are first stored. Next, the classification of a new unclassified record is performed by **comparing it to the most similar records in the training set.**

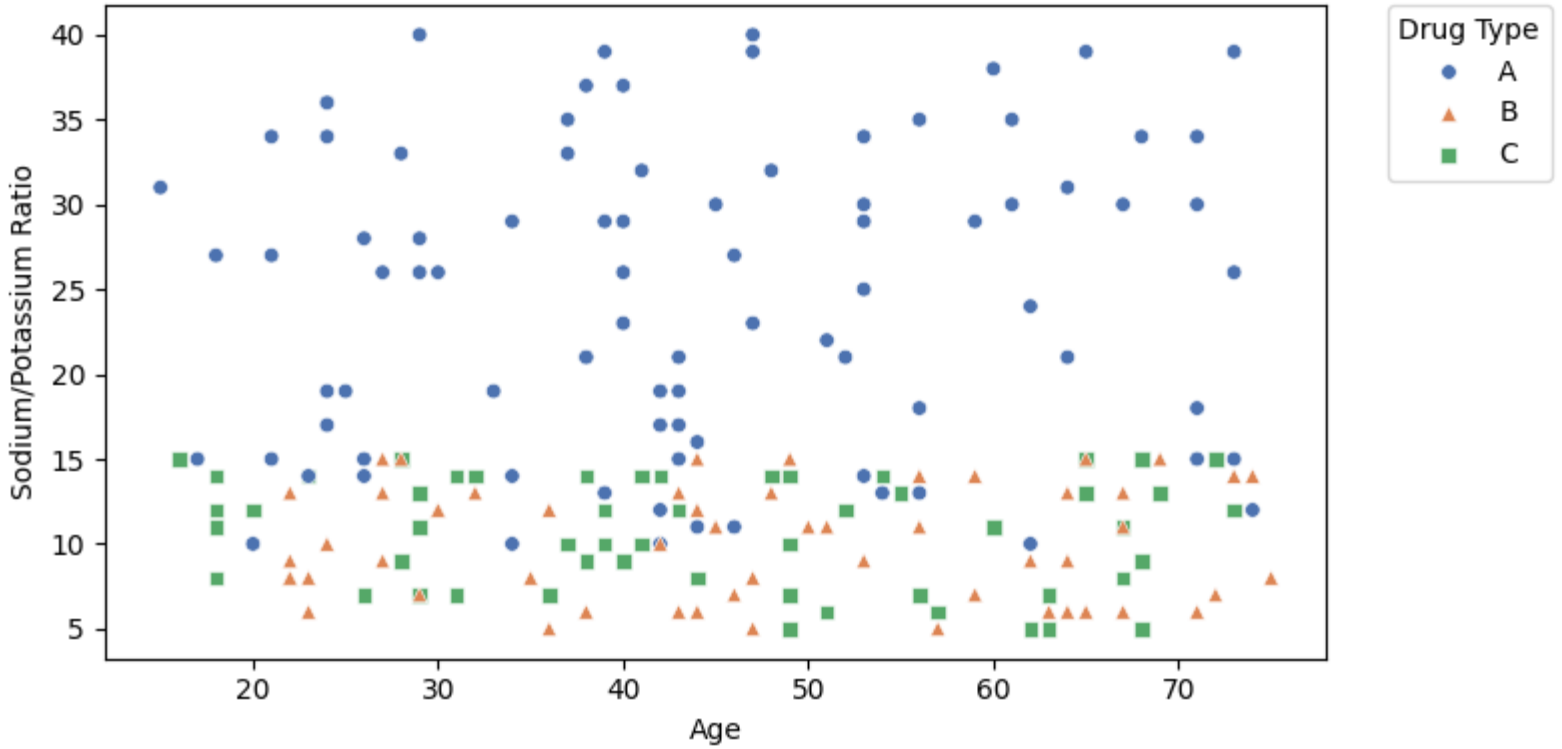
Example: Suppose we are interested in classifying the type of drug a patient should prescribed. The training set consists of **200 patients** with **sodium/potassium (Na/K) ratio**, ***age*** and ***drug prescribed***.

Our aim is to classify the type of drug a new patient should be prescribed that is:

- **Patient 1**: 40 years old with a Na/K ratio of 30.5.
- **Patient 2**: 28 years old with a Na/K ratio of 9.6.
- **Patient 3**: 61 years old with a Na/K ratio of 10.5.

UvA

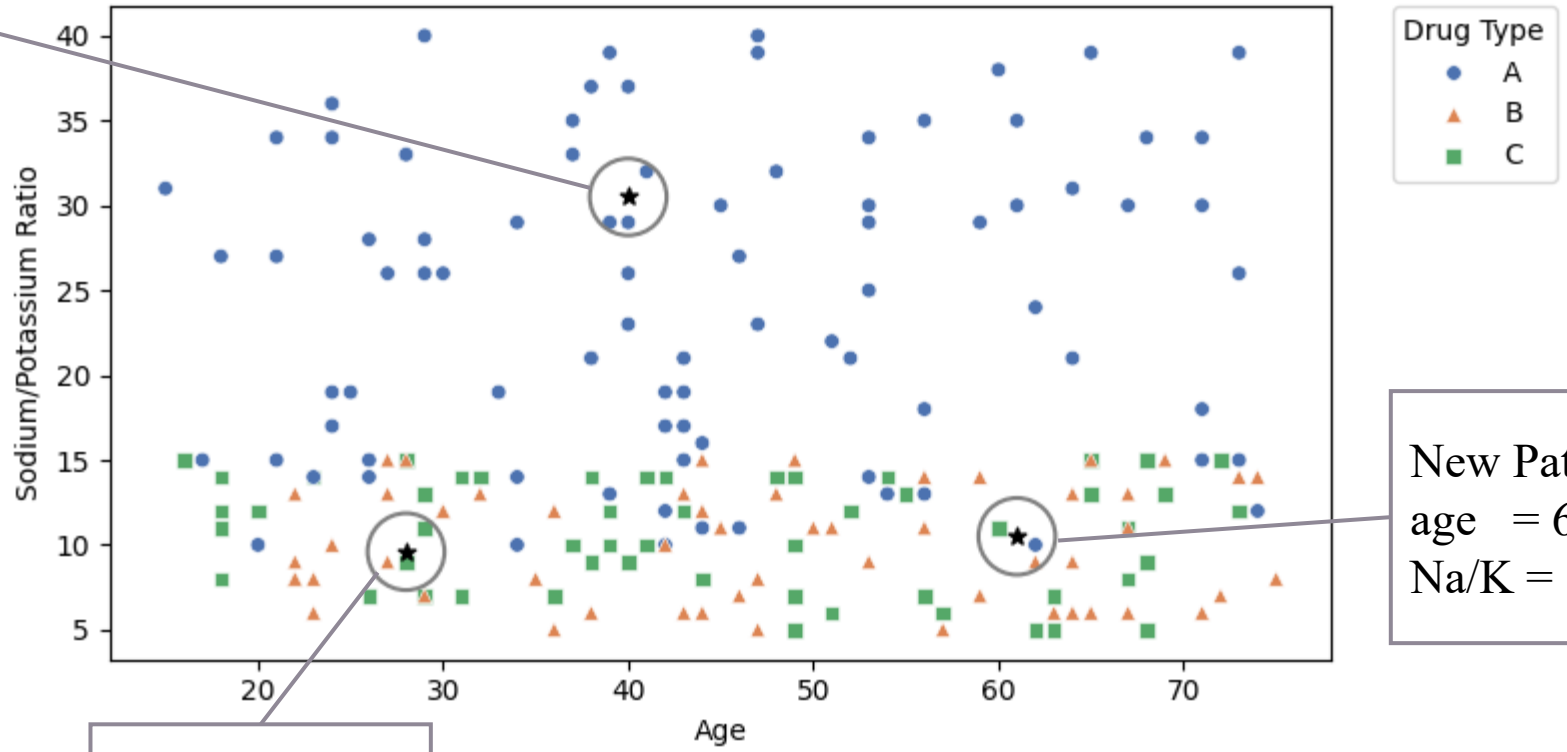
k-Nearest Neighbor Algorithm



UvA

k-Nearest Neighbor Algorithm

New Patient 1
age = 40
Na/K = 30.5



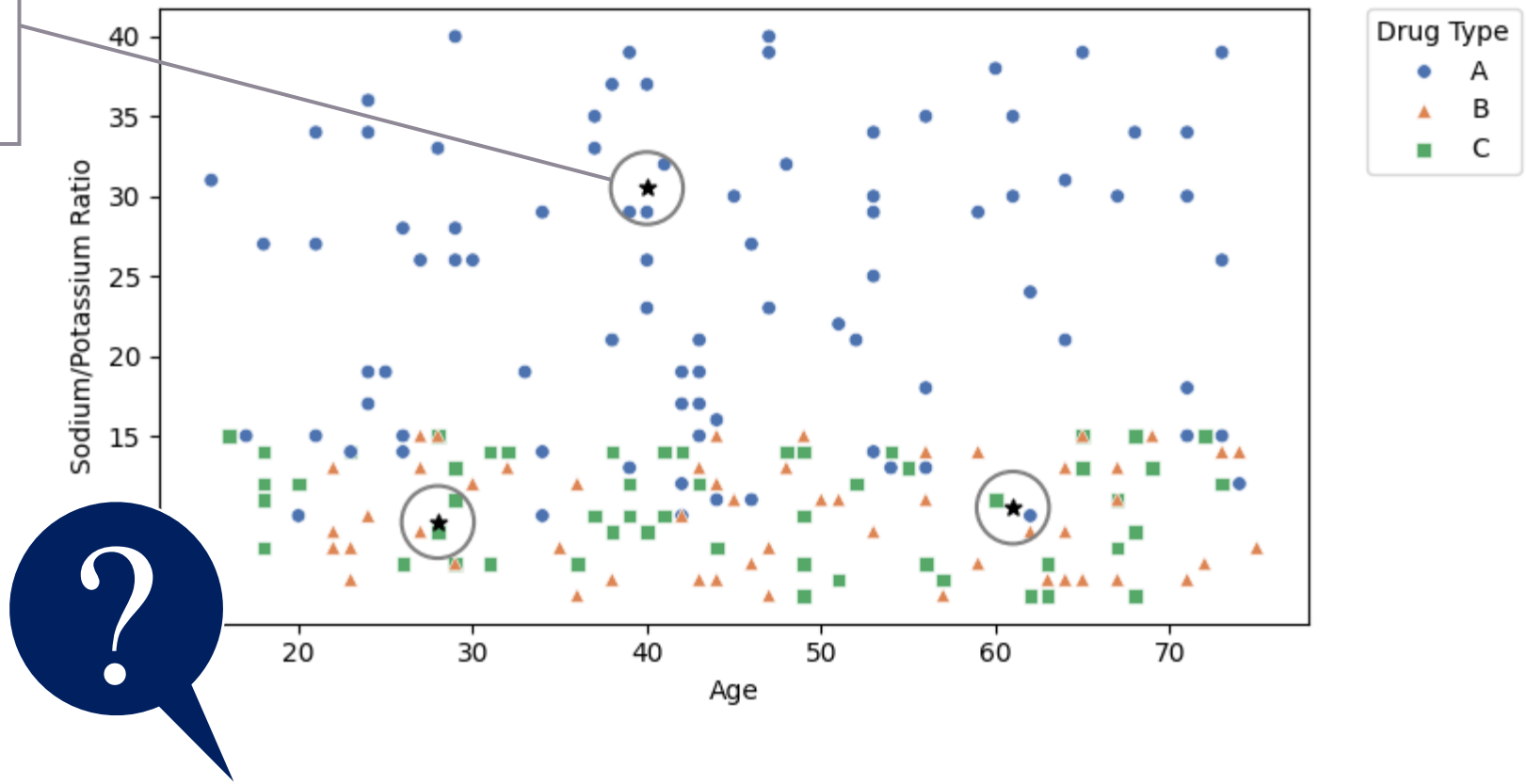
New Patient 3
age = 61
Na/K = 10.5

New Patient 2
age = 28
Na/K = 9.6

UvA

k-Nearest Neighbor Algorithm

New Patient 1
age = 40
Na/K = 30.5



Which drug should **Patient 1** be prescribed?

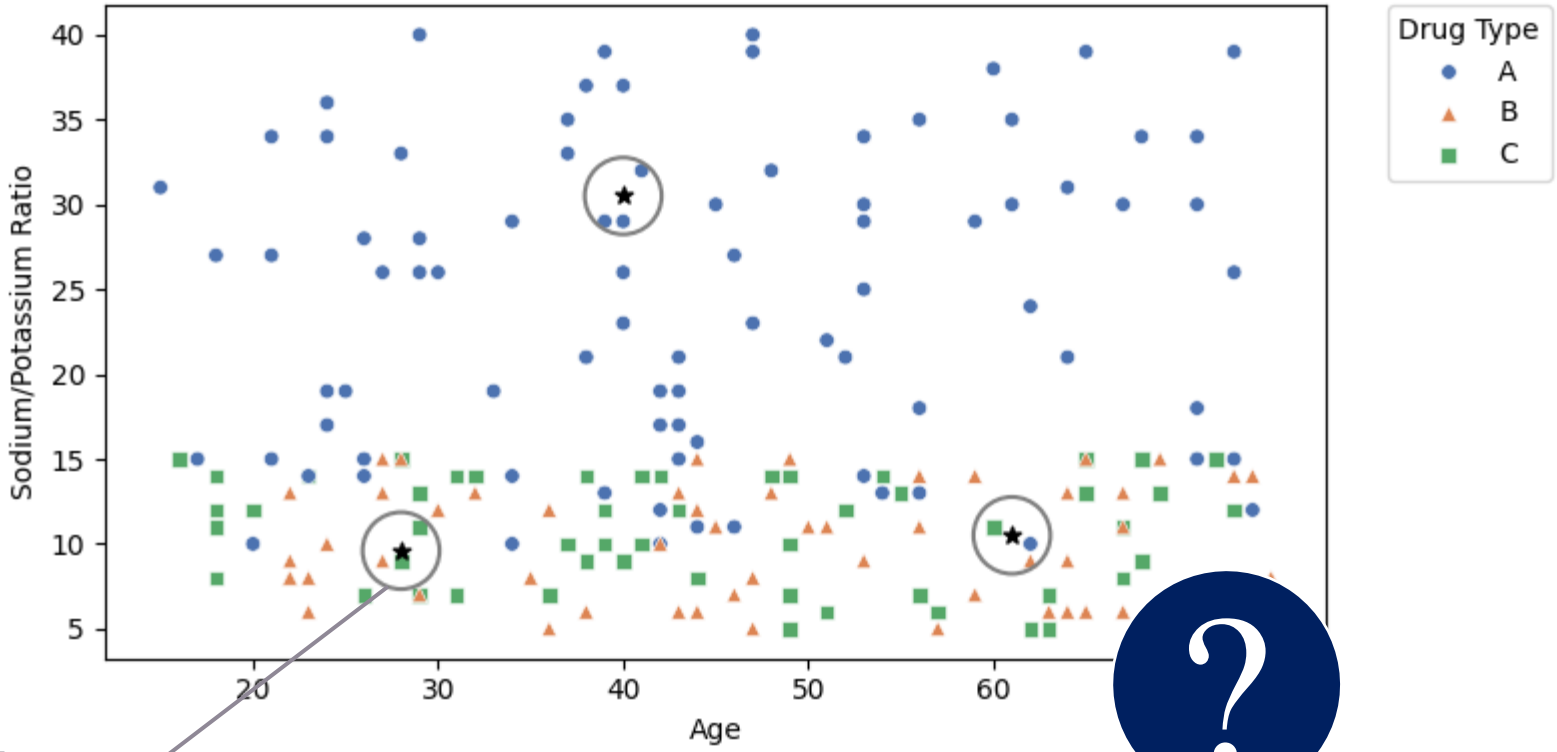
UvA

k-Nearest Neighbor Algorithm

Patient 1

Since Patient 1's profile places them in the scatter plot near patients prescribed drug A, we classify Patient 1 as drug A. All points near Patient 1 are prescribed drug A, making this a straightforward classification.

k-Nearest Neighbor Algorithm



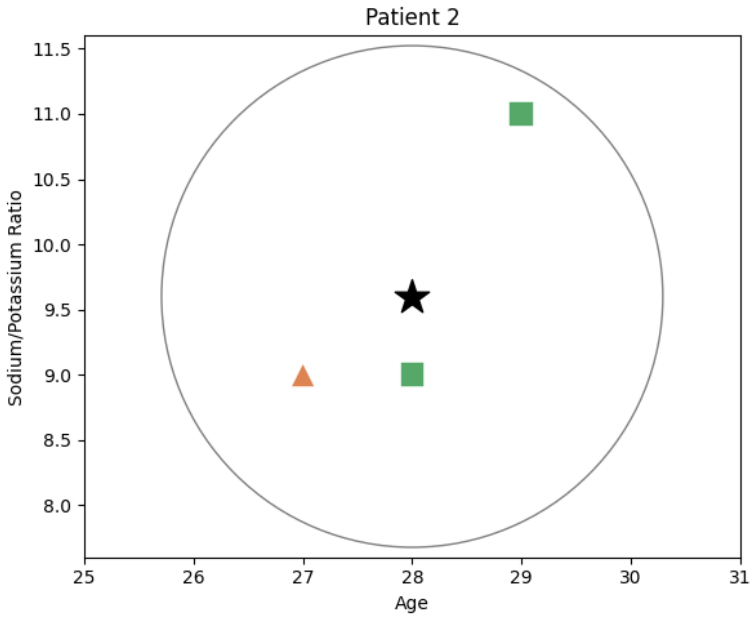
New Patient 2
age = 28
Na/K = 9.6

What can we say about **patient 2**?

UvA

k-Nearest Neighbor Algorithm

Close-up of three nearest neighbors to **New Patient 2.**



Patient 2 illustrates sensitivity to the choice of *k*

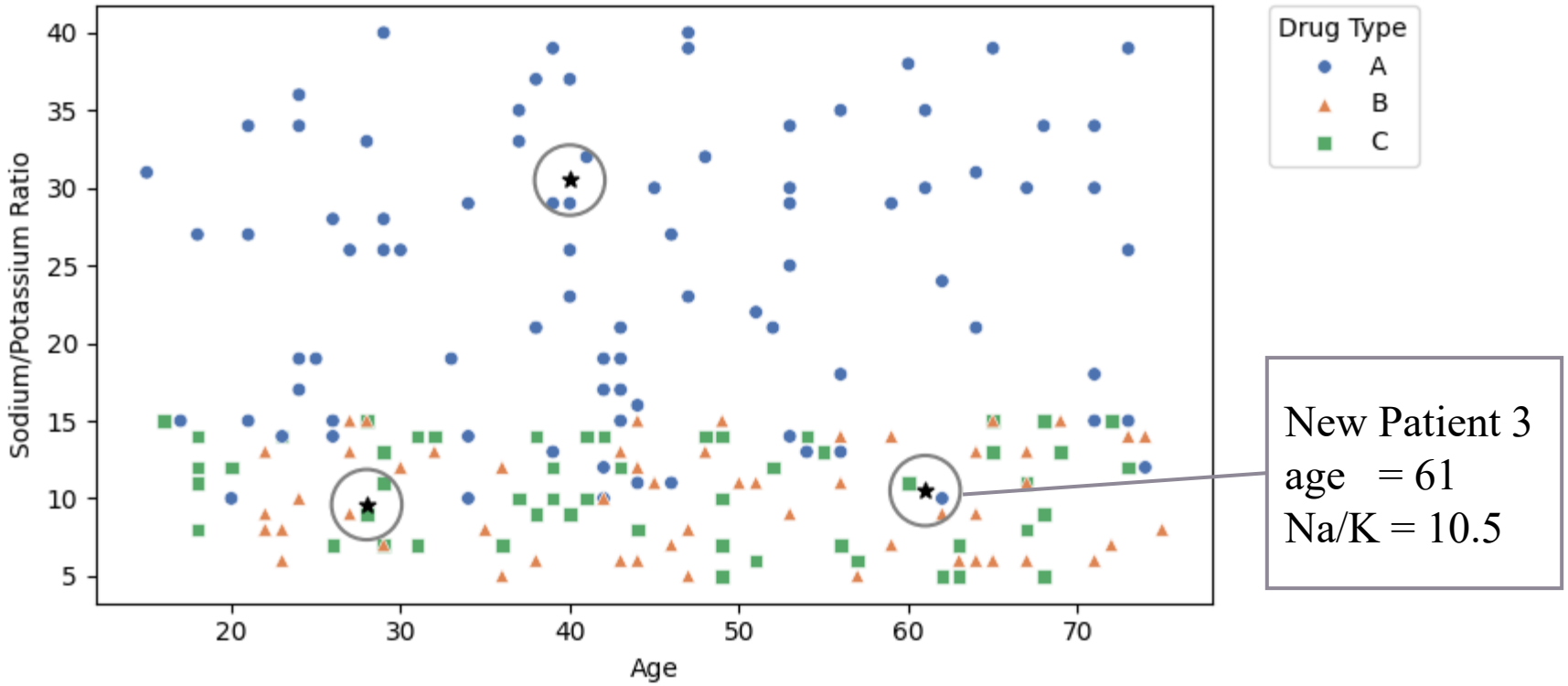
k=1: Highly sensitive to local variation.

k=2: The result is a tie.

Note: The classification of Patient 2 differs based on the value chosen for *k*.

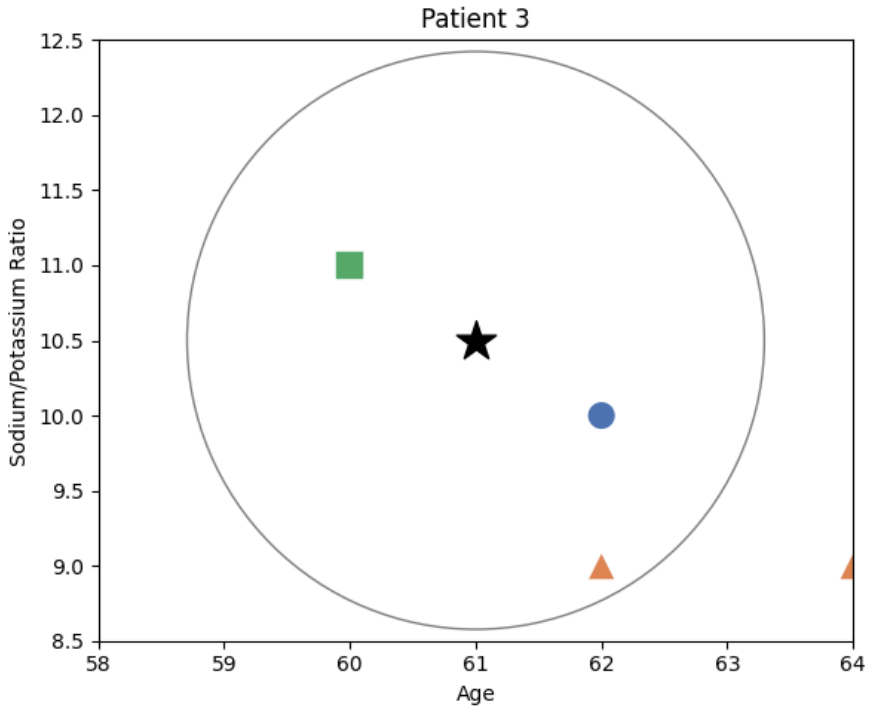
UvA

k-Nearest Neighbor Algorithm



k-Nearest Neighbor Algorithm

Close-up of three nearest neighbors to **New Patient 3.**



- With $k = 1$ **or** $k = 2$, Patient 3 is closest to both red and blue points. So, voting does not help.
- With $k = 3$, voting does not help since each of the nearest training points has a different classification.

UvA

How to Measure Similarity?

Suppose you're building a computer vision system. Given two images, what measurable features could you use to quantify their similarity?

Similarity is hard to define, but… “We know it when we see it”

Source: https://knowyourmeme.com/photos/1090578-puppy-or-bagel

UvA

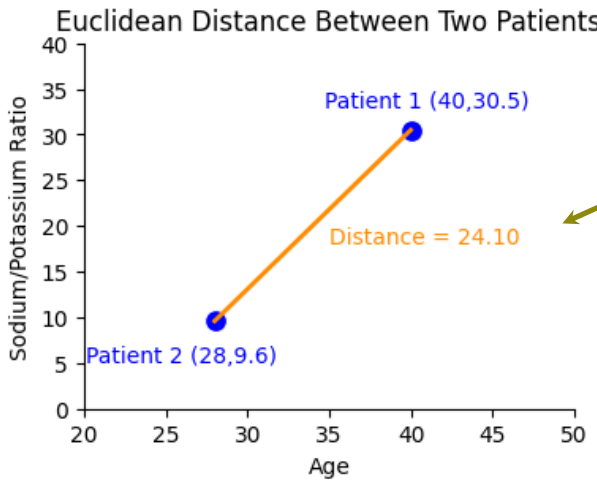
Similarity Measures

The **Euclidean Distance** is commonly used to measure distance

$$d(x,y) = \sqrt{\sum_i (x_i - y_i)^2}$$

where $\boldsymbol{x}_i = x_1, x_2, \ldots, x_m$ *and* $\boldsymbol{y}_i = y_1, y_2, \ldots, y_m$ *represent the m attribute values of two records.*

Example: Suppose Patient 1 is 40 years old and has a Na/K ratio = 30.5, and Patient 2 is 28 years old and has a Na/K ratio = 9.6. What is the distance between these instances?



$$d(x,y) = \sqrt{\sum_i (x_i - y_i)^2}$$

$$= \sqrt{(40-28)^2 + (30.5-9.6)^2}$$

$$= 24.1$$

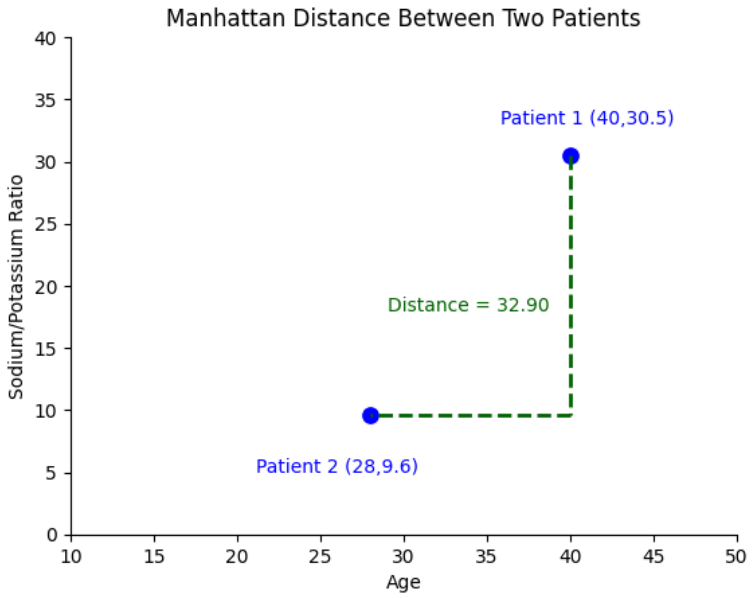
UvA

Similarity Measures

The **Manhattan Distance** is another commonly used to measure distance

$$d(x, y) = \sum_i |x_i - y_i|$$

where $\boldsymbol{x}_i = x_1, x_2, \ldots, x_m$ *and* $\boldsymbol{y}_i = y_1, y_2, \ldots, y_m$ *represent the m attribute values of two records.*



$$d(x, y) = \sum_i |x_i - y_i|$$

$$= 40 - 28 + 30.5 - 9.6$$

$$= 32.9$$

This is used when:

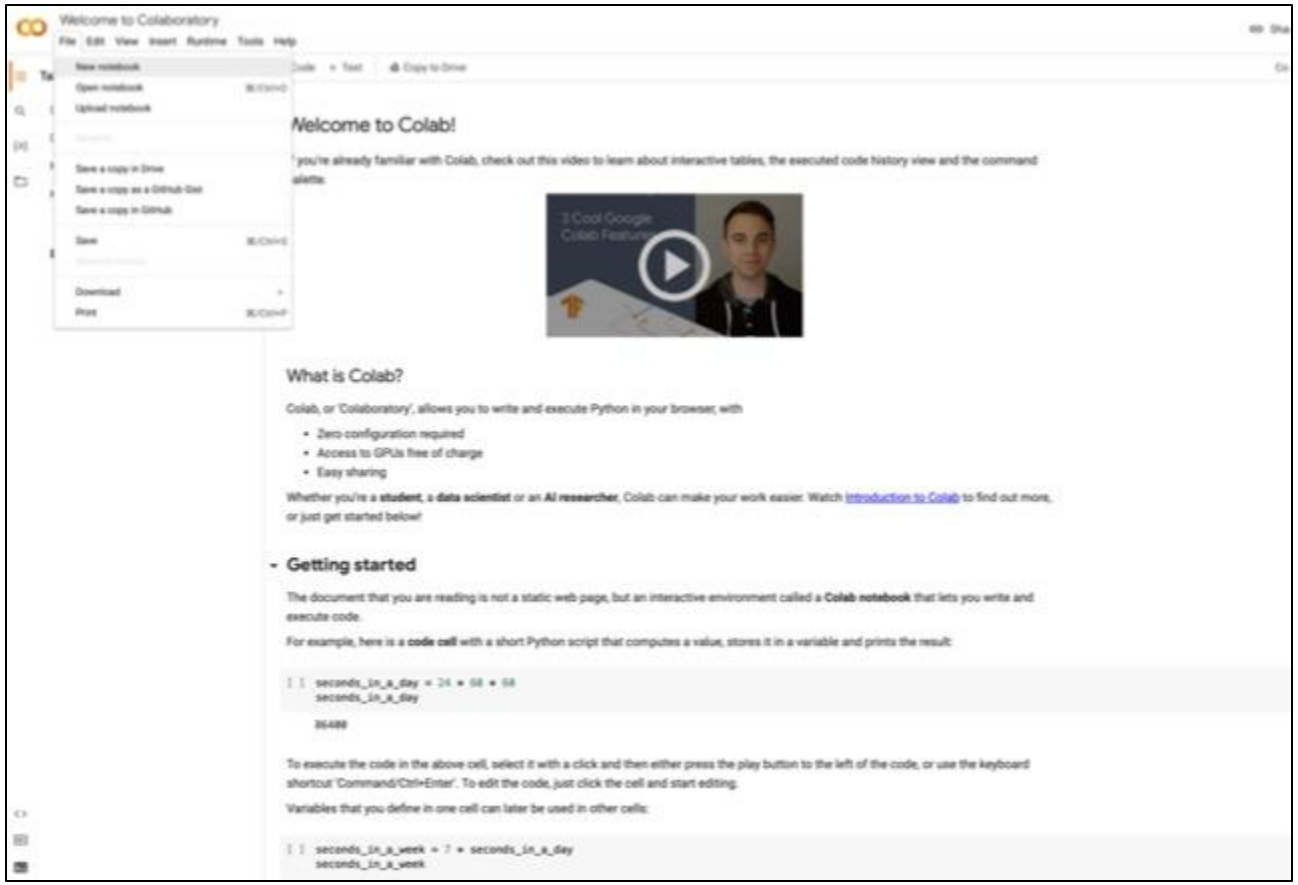
- Differences should be treated independently per dimension
- Movement is constrained to axis-aligned paths (like city blocks)
- You want a metric that is less sensitive to large outliers

UvA

Jupyter Notebook

- You are not expected to memorize this code. Today we're learning how machine learning works and how to experiment with it, not Python syntax.

Google Colab



colab.research.google.com

UvA

Advantages and Disadvantages of *k*-Nearest Neighbors

Advantages:

- No training phase required
- Easy to implement and interpret
- Non-parametric: no assumptions about distribution of the data

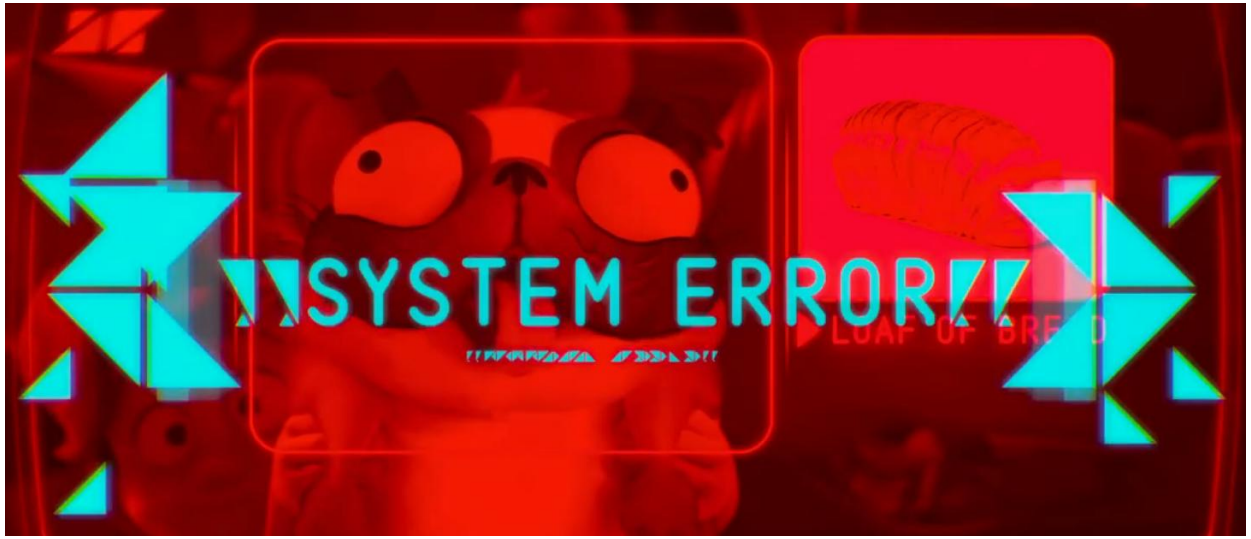
Disadvantages:

- Hard to determine 'correct' k
- Computationally inefficient with large and multi-dimensional datasets
- Sensitive to outliers
- Need to normalize data

Evaluation

Examples of evaluation methods:

- Confusion Matrix
- Metrics:
  - Accuracy
  - Precision
  - Recall
- ROC / AUC



Confusion Matrix

A confusion matrix shows the number of correct and incorrect predictions made by the classification model compared to the actual outcomes (target value) in the data. The matrix is $N \times N$, where N is the number of target values (classes). The following table displays a 2×2 confusion matrix for two classes (Negative and Positive):

		Predicted Values: Negative	Predicted Values: Positive
Actual Values	Negative	True Negative (TN)	False Positive (FP)
Actual Values	Positive	False Negative (FN)	True Positive (TP)

Like the hypothesis testing, you may recall the two errors we defined as type-I and type-II.

UvA

Metrics

- Accuracy: Measures the proportion of all predictions that are correct.

$$\textbf{Accuracy} = \frac{TP+TN}{TP+FP+TN+FN}$$

- Precision: Measures the proportion of predicted positives that are actually positive.

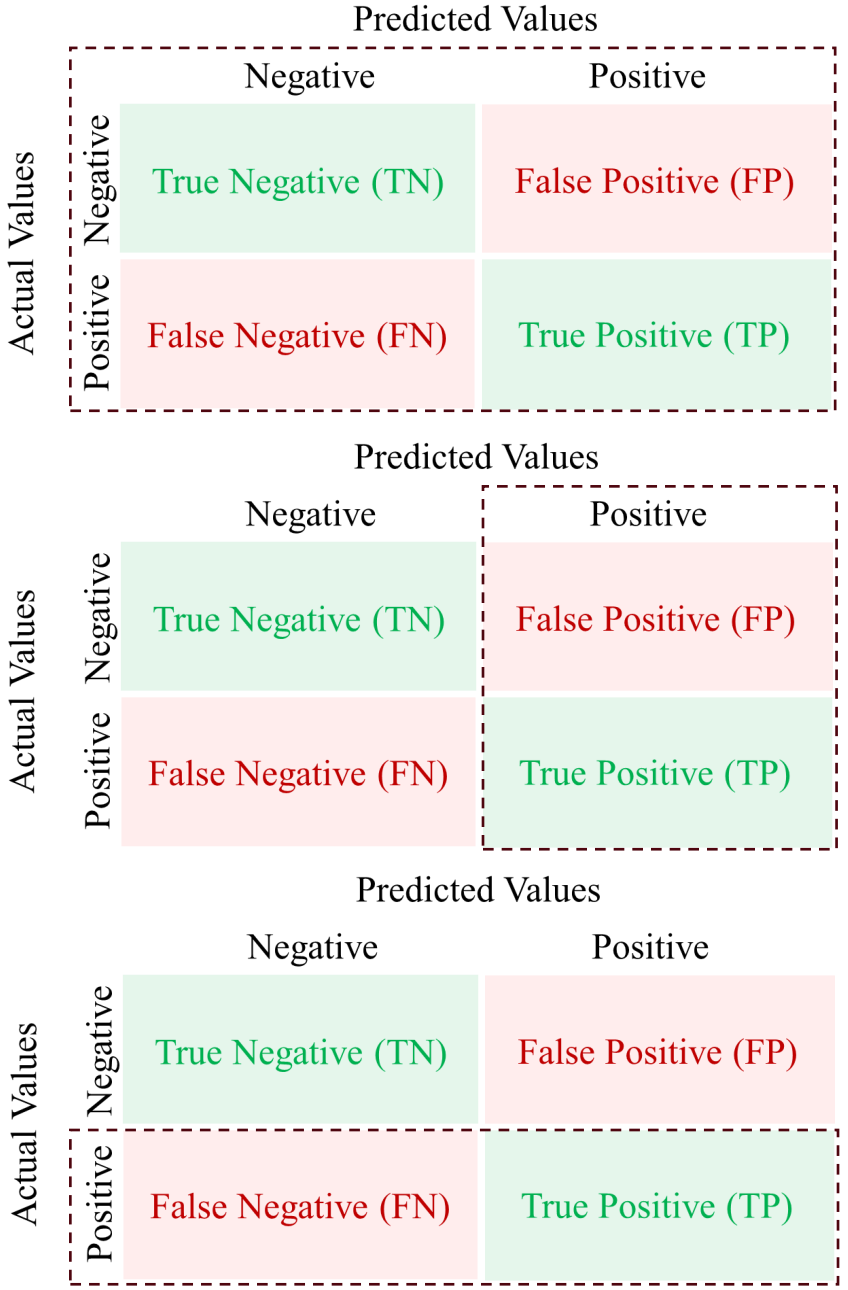
$$\textbf{Precision} = \frac{TP}{TP + FP}$$

Important when the cost of false positives is high (e.g., important mail classified as spam or false medical diagnoses leading to unnecessary treatments and emotional distress)

- Recall: Measures the proportion of actual positives that are correctly identified.

$$\textbf{Recall} = \frac{TP}{TP + TN}$$

Important when cost of false negatives is high (e.g., missing fraud)



ROC curve

An ROC curve (receiver operating characteristic curve) is a graph showing the performance of a binary classification model at all classification thresholds. This curve plots two parameters:

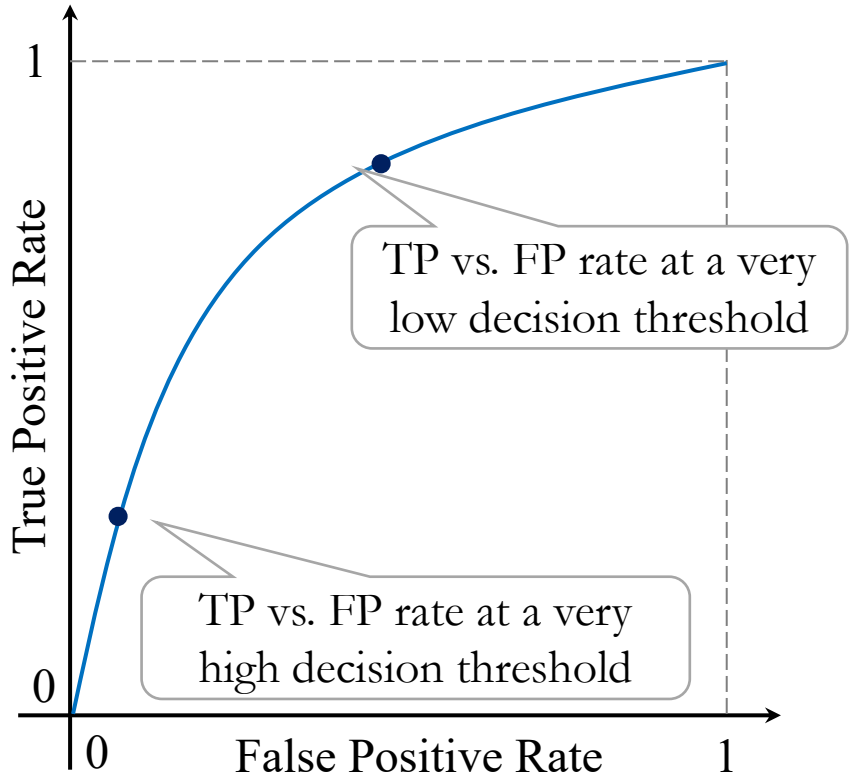
FPR: Out of all *actual negatives*, how many did we incorrectly label as positive?

$$\textbf{False Positive Rate} = \frac{\textbf{FP}}{\textbf{FP} + \textbf{TN}}$$

TPR: Out of all *actual positives*, how many did we correctly catch?

$$\textbf{True Positive Rate} = \frac{\textbf{TP}}{\textbf{TP} + \textbf{FN}}$$

ROC curve



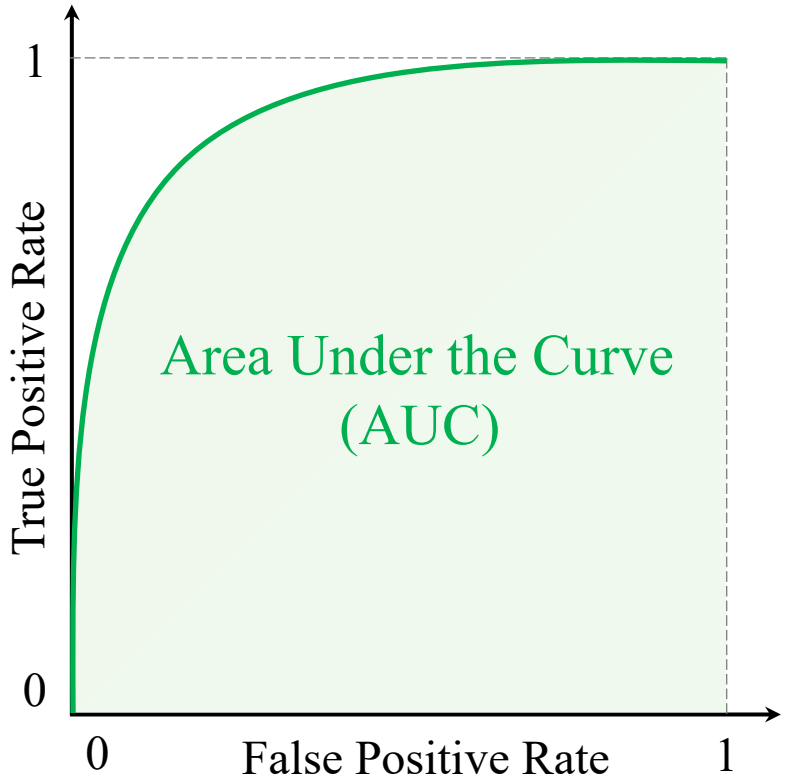
$$\textbf{False Positive Rate} = \frac{\textbf{FP}}{\textbf{FP} + \textbf{TN}}$$

$$\textbf{True Positive Rate} = \frac{\textbf{TP}}{\textbf{TP} + \textbf{FN}}$$

UvA

Area Under the Curve (AUC)

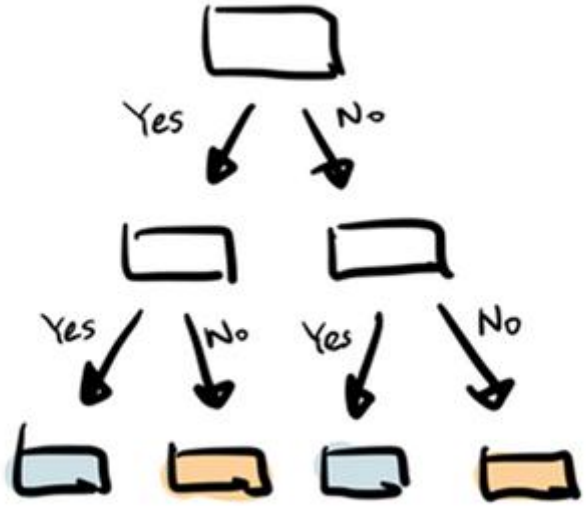
The area under the ROC curve is often a measure of the quality of the classification models. A random classifier has an area under the curve of 0.5, while AUC for a perfect classifier is equal to 1. In practice, most of the classification models have an AUC between 0.5 and 1.



UvA

Decision Trees

- Employ a divide-and-conquer method.
- Decision tree builds tree branches in a hierarchy approach.
- Each branch can be considered as an if-else statement.
- The branches develop by partitioning the dataset into subsets based on most important features.
- Final classification happens at the leaves of the decision tree.

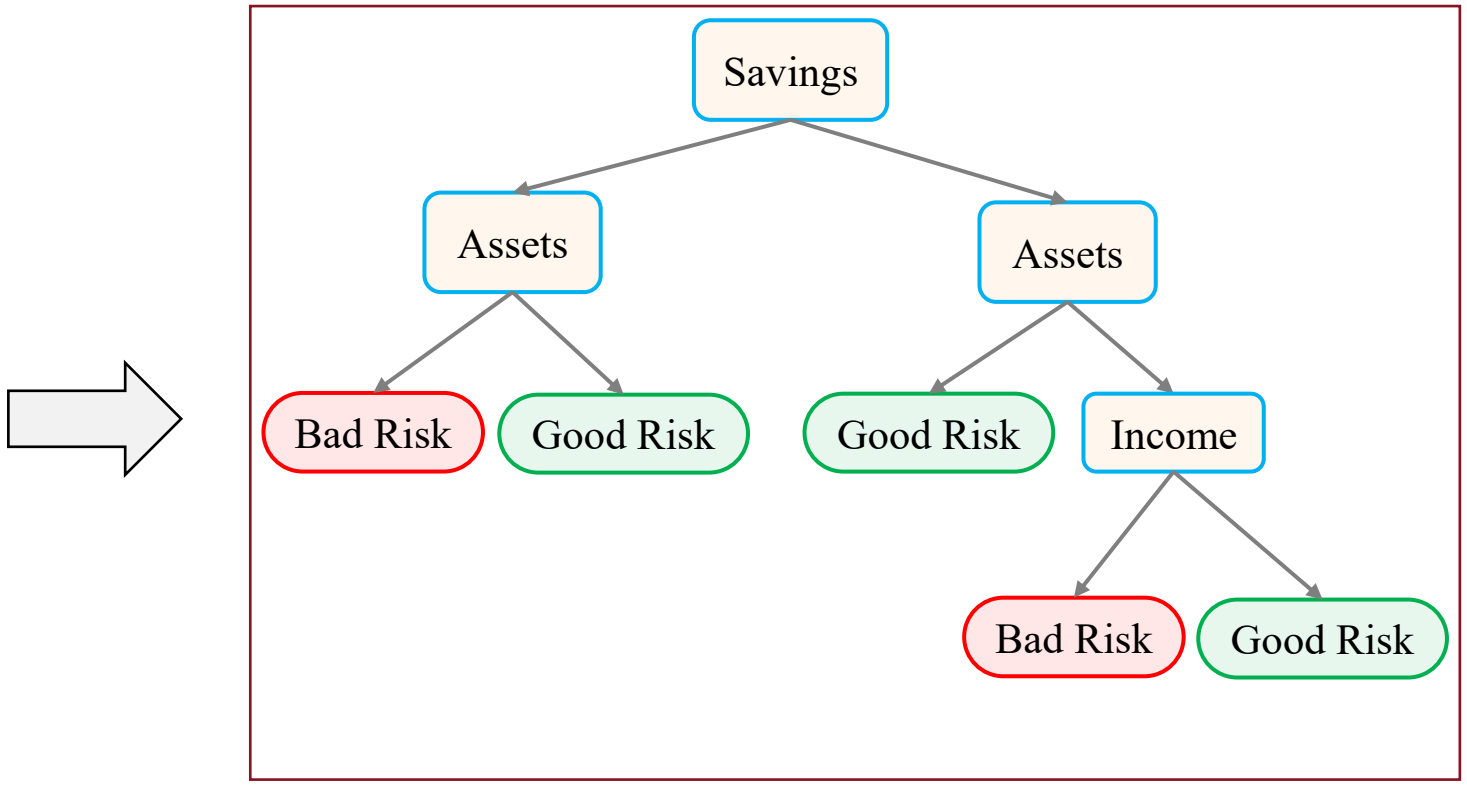


UvA

Decision Trees

Features Target

	Savings	Assets	Income	Label
0	Low	Low	Low	Bad Risk
1	Low	Low	High	Bad Risk
2	Low	High	Low	Good Risk
3	Low	High	High	Bad Risk
4	Low	High	High	Good Risk
5	Low	Low	Low	Bad Risk
6	High	Low	Low	Good Risk
7	High	Low	High	Good Risk
8	High	High	Low	Bad Risk
9	High	High	High	Good Risk
10	High	High	Low	Good Risk
11	Low	Low	High	Bad Risk
12	High	Low	Low	Good Risk
13	High	High	High	Good Risk
14	Low	High	Low	Good Risk
15	Low	Low	High	Bad Risk
16	High	Low	High	Good Risk
17	Low	High	High	Good Risk

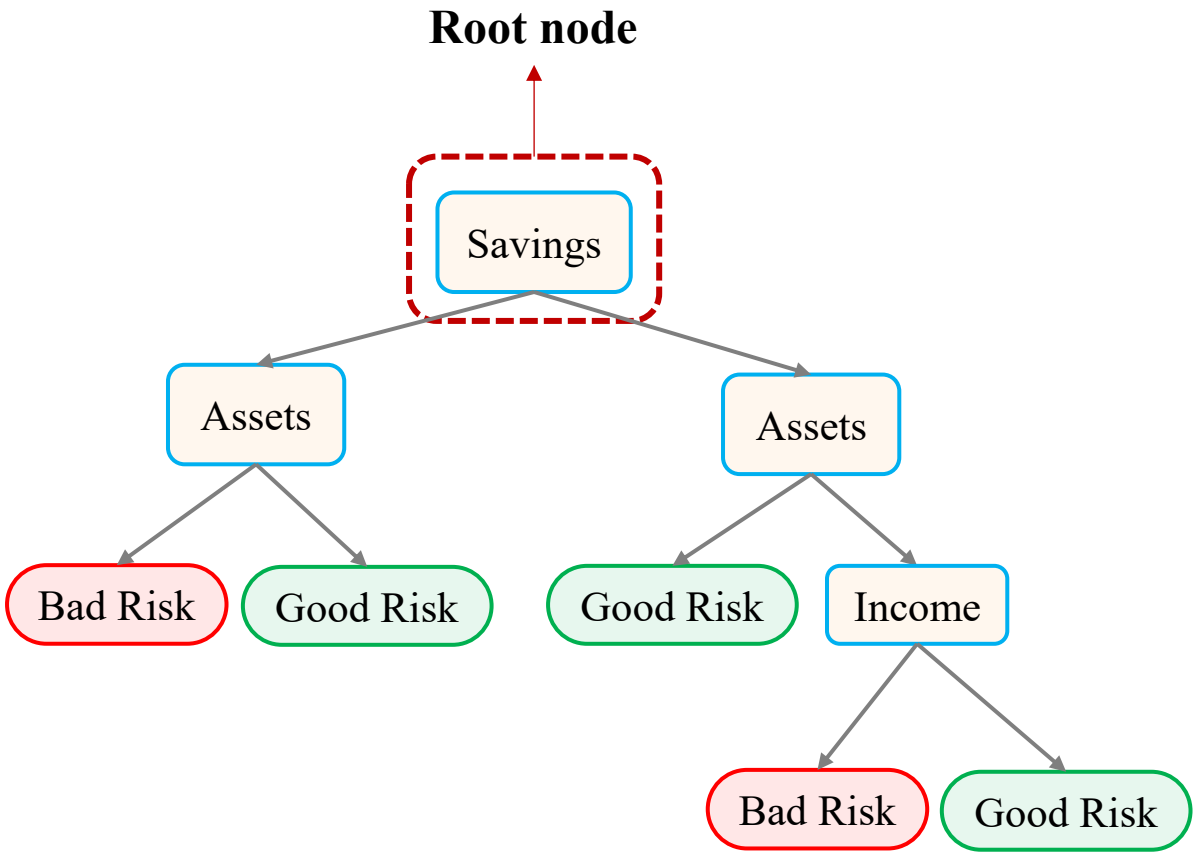


UvA

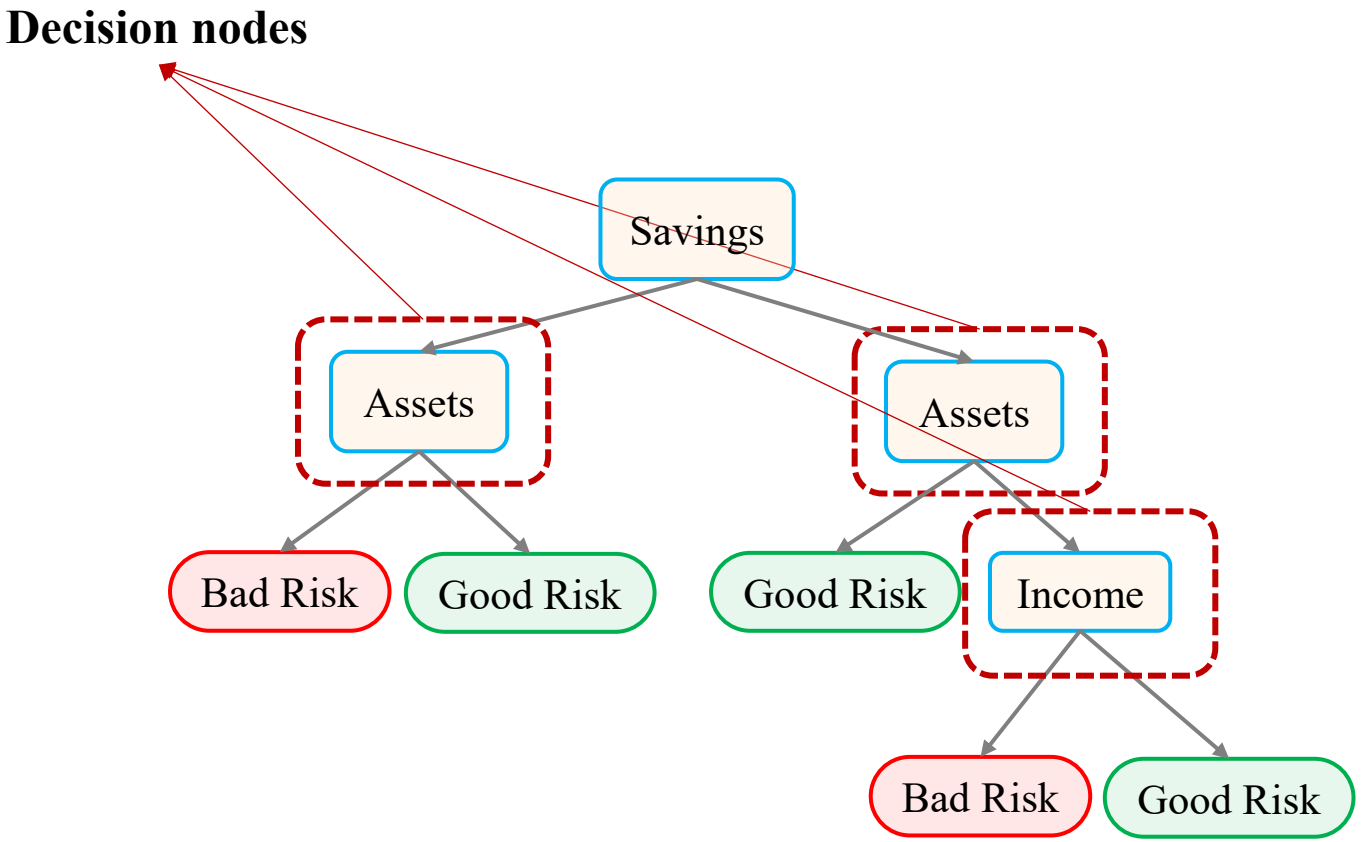
Decision Tree Classifier

- The **target variable must be categorical** with pre-classified values included in the training set.
- A decision tree is a collection of **decision nodes**, connected by **branches**, extending downward from the **root node** to terminating **leaf nodes.**
- Beginning with the root node, attributes are tested at the decision nodes, and each possible outcome results in a branch.
- Each branch leads to a decision node or to a terminating leaf node.

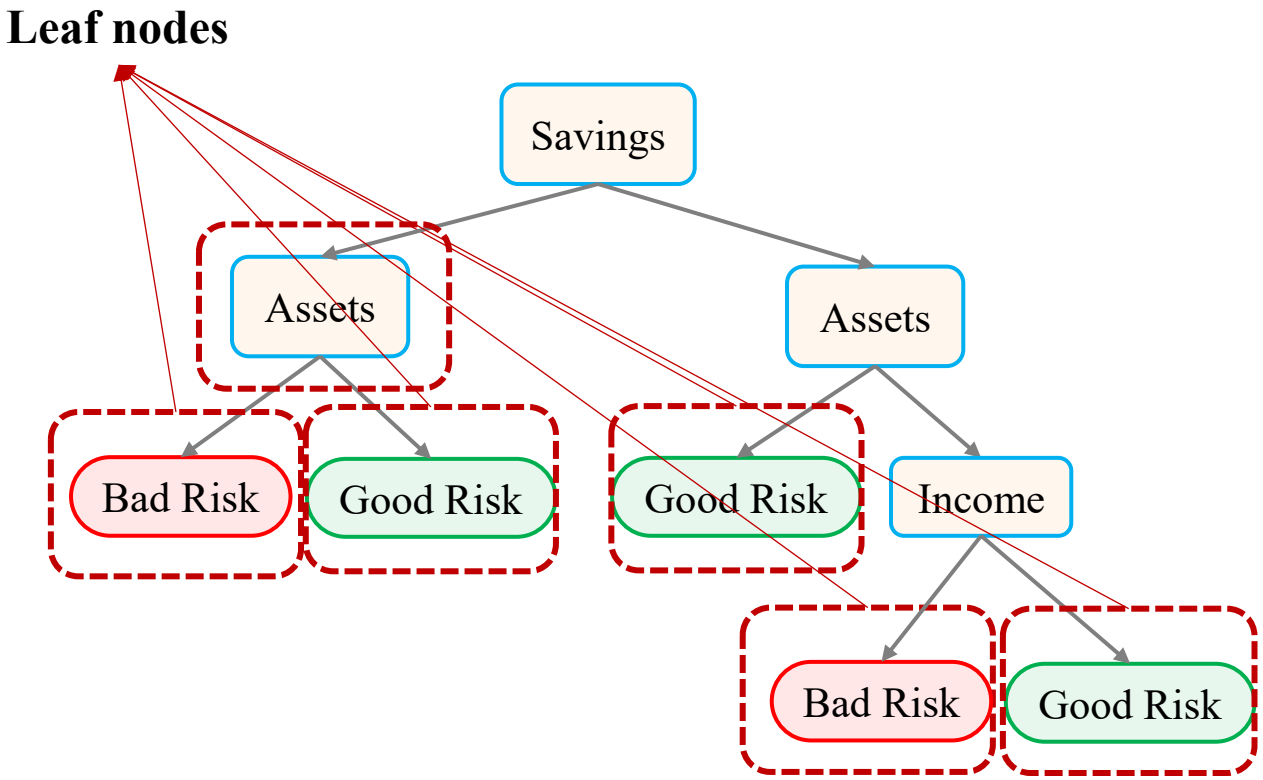
Decision Trees



Decision Trees

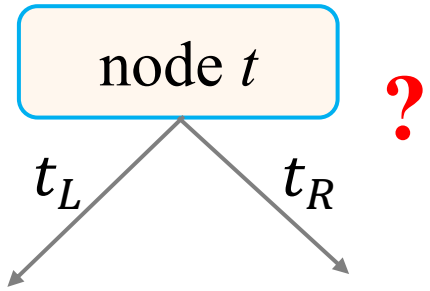


Decision Trees



Decision Tree Algorithm

- Splits at decision nodes are strictly binary, resulting in two branches.
- The algorithm grows the tree by evaluating all predictor variables and choosing the optimal split according to the largest reduction in impurity.
- Repeats this process recursively on each resulting subset until no further split improves purity or a stopping condition is reached.



Splitting criteria

- Gini impurity
  - Measures probability of misclassification
  - Interprets how often a random label would be wrong
  - Used as default in scikit-learn
- Entropy: Measure of uncertainty
  - Measures uncertainty in the data
  - From information theory
  - Based on information gain

- Key idea:
  - Both measure node impurity (how mixed classes are)
  - Both aim to create more pure child nodes after splits
  - Usually produce very similar trees and results
  - Gini is slightly faster to compute
  - Entropy has a stronger theoretical foundation

Jupyter Notebook

Tree pruning parameters

max_depth: Limits how deep the tree can grow, reducing complexity and helping prevent overfitting

min_samples_split: Sets the minimum number of samples required before a node can be split, controlling how easily the tree branches

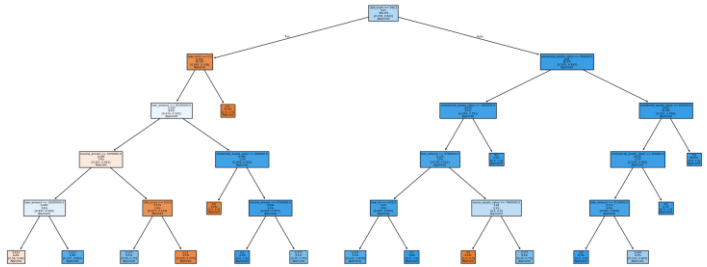
min_samples_leaf: Defines the minimum number of samples a leaf node must have, preventing overly specific (noisy) leaves

max_leaf_nodes: Restricts the total number of leaf nodes in the tree, directly limiting overall model size

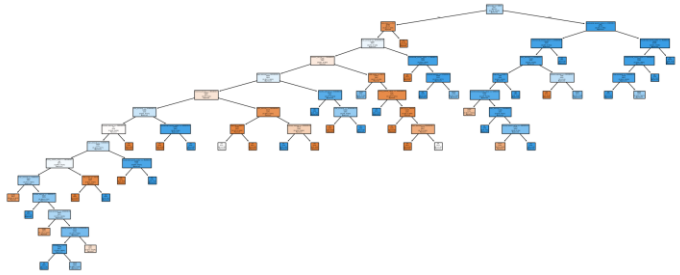
ccp_alpha: Controls post-pruning strength using cost-complexity pruning; higher values simplify the tree by removing weak branches

Pruning

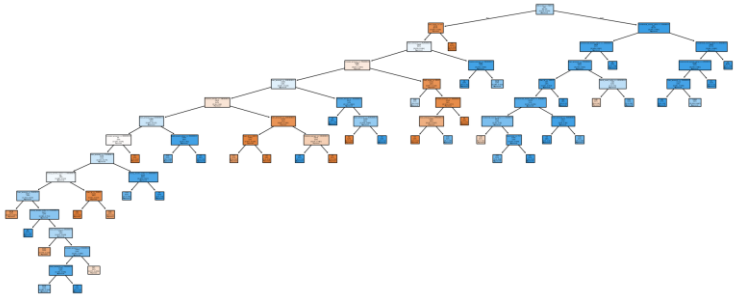
max_depth=5



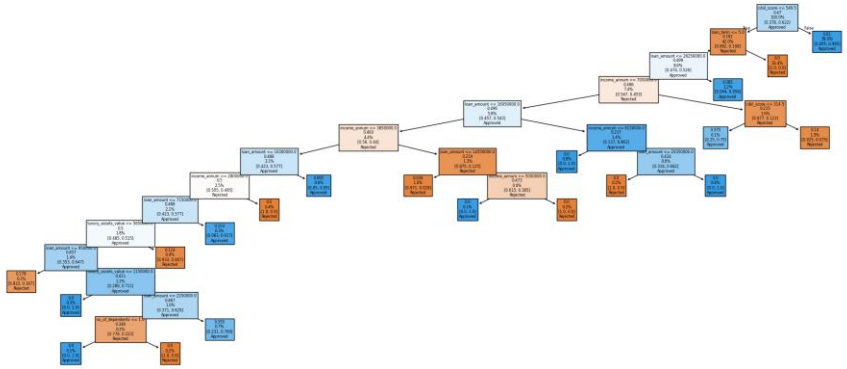
min_samples_split=10



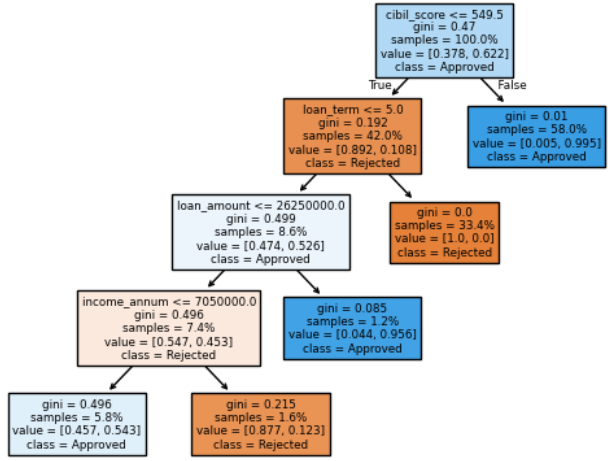
min_samples_leaf=10



max_leaf_nodes=20



ccc_alpha=0.004



UvA

Decision Tree

Advantages:

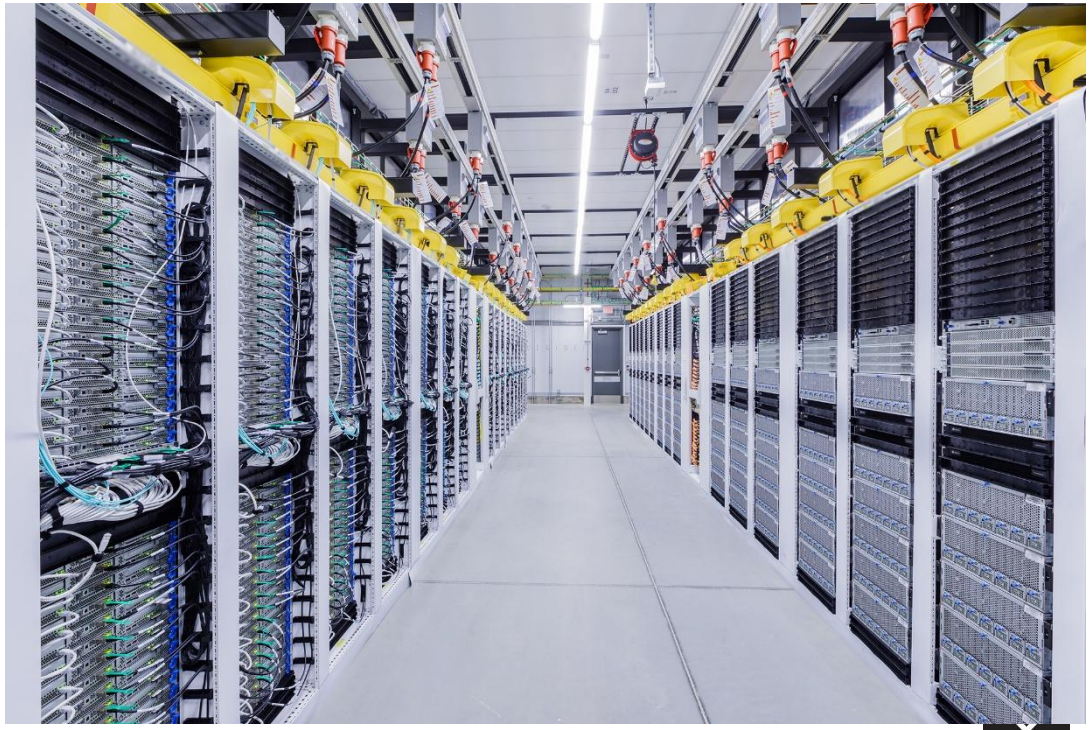
- Intuitive and easy to explain
- Do not require normalization of data
- Robust to missing data and outliers
- Non-parametric (no assumptions about underlying distributions)

Disadvantages:

- Need to control tree complexity (prone to overfitting)
- New or changed data can affect the structure of the decision tree considerably
- Can require significant resources to train
- Greedy algorithm: can result in non-optimal solutions

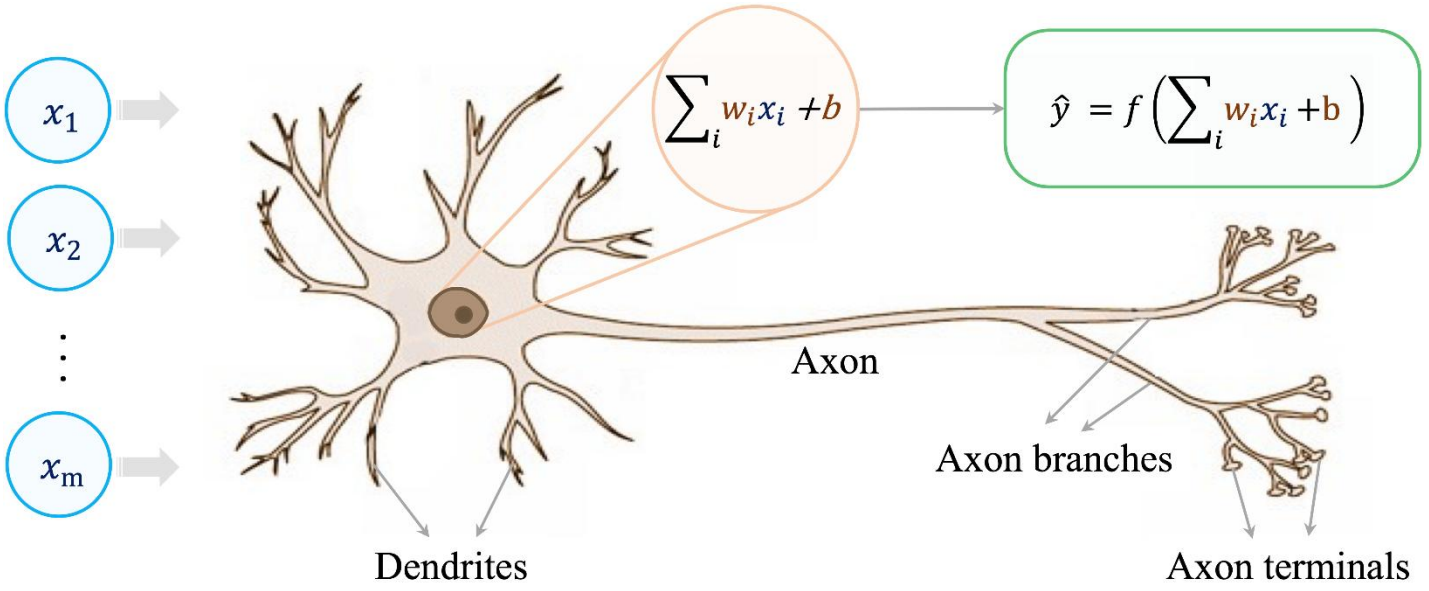
Neural Networks

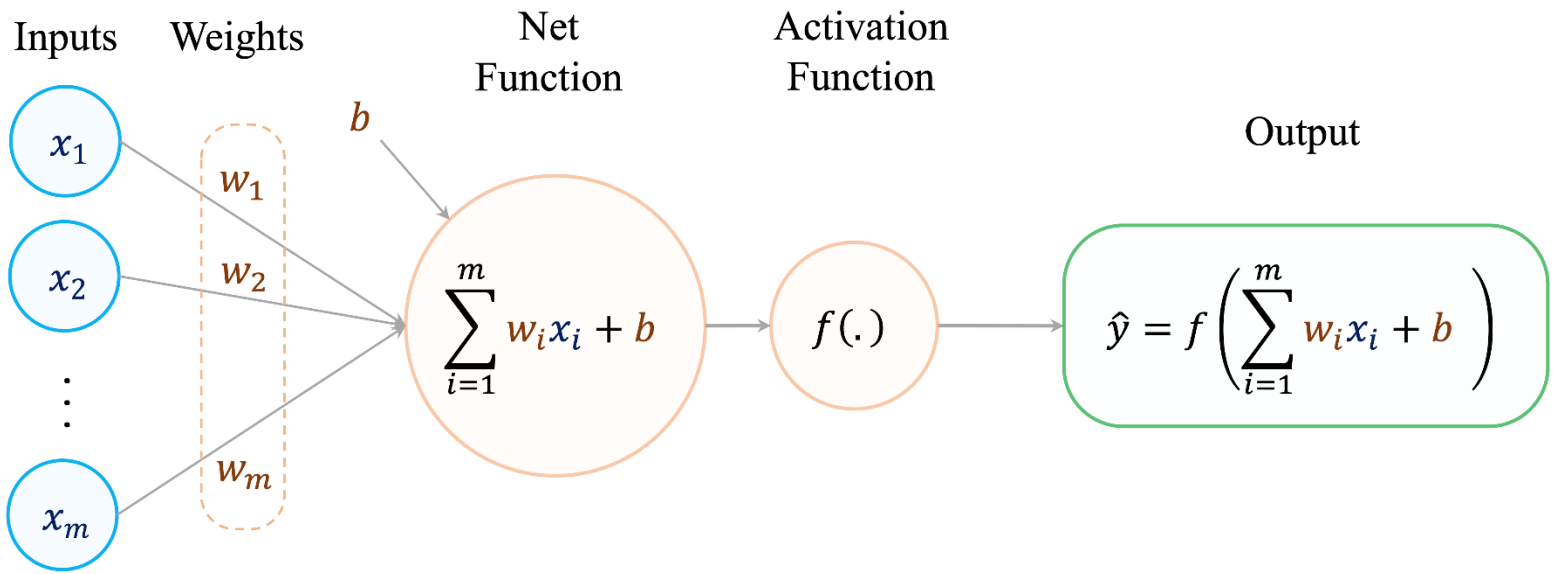
- Inspired by complex learning systems recognized in brain
  - *Single neuron has simple structure*
  - *Interconnected sets of neurons perform complex learning tasks*
- Artificial Neural Networks attempt to replicate non-linear learning found in nature
- The ability of neural networks to discover patterns allows them to far surpass traditional methods in computer science, making them viable for use in research and business



UvA

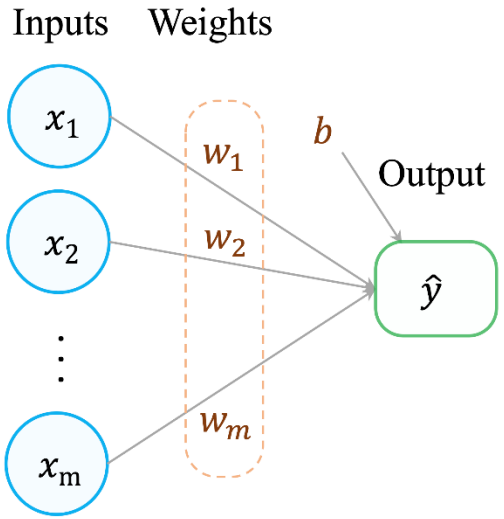
Neural Networks



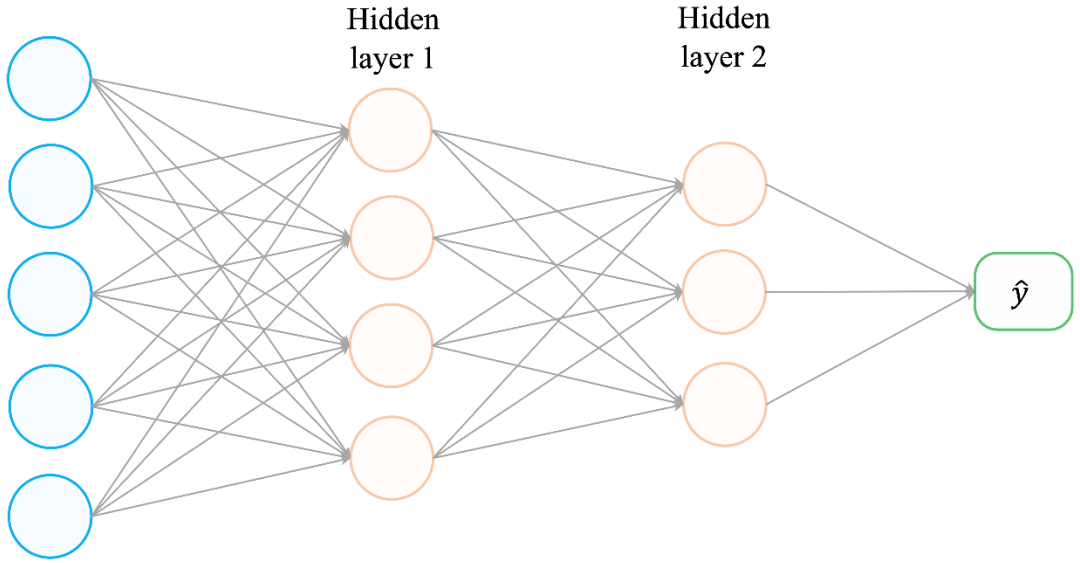


UvA

Simple to Advanced Neural Networks



It's similar to logistic regression models

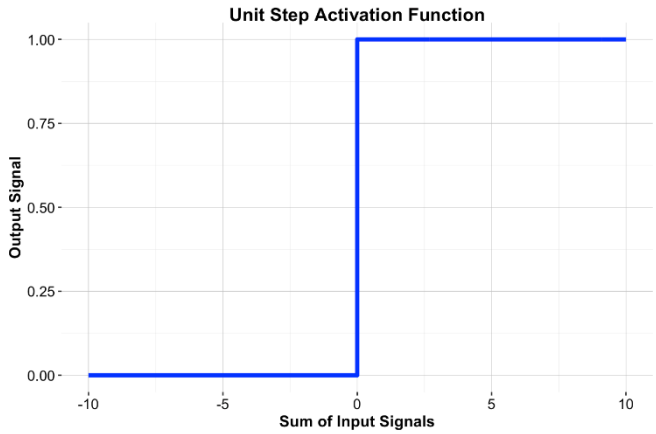


UvA

Activation Function

One option for *f* could be a simple Threshold function:

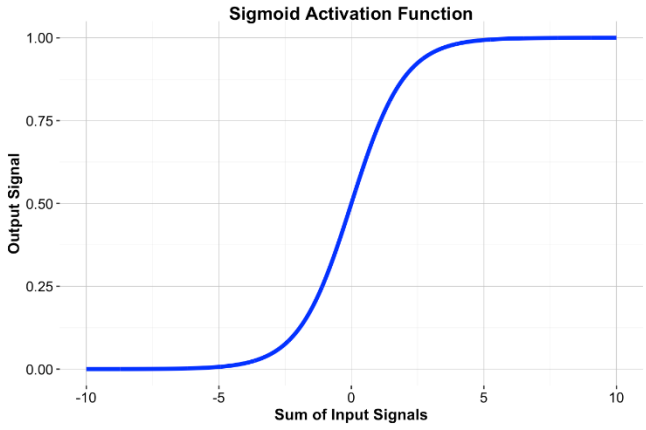
$$f(x) = \begin{cases} 1 \; if \; x \geq 0 \\ 0 \; if \; x < 0 \end{cases}$$



Sigmoid Function: combines nearly linear, curvilinear, and nearly constant behavior depending on input value:

$$f(x) = \frac{1}{1 + e^{-x}}$$

This function returns values [0, 1].



UvA

Activation Functions in Neural Networks

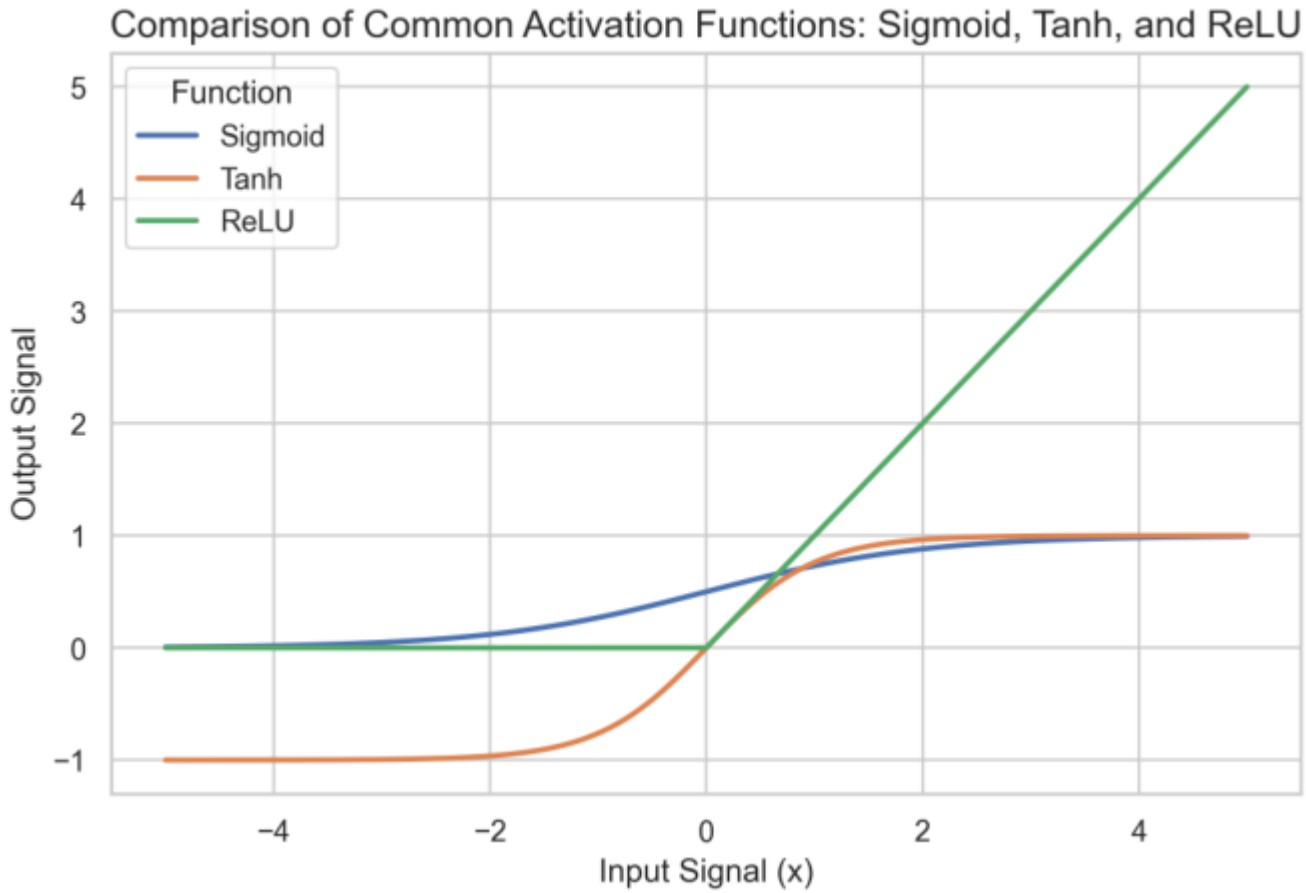
While the sigmoid function was foundational in early neural networks, modern architectures benefit from activation functions that offer faster convergence and improved gradient flow. The most widely used alternatives are

- *Hyperbolic Tangent (tanh):* Like sigmoid, but outputs values between -1 and 1, making it zero-centered and often better suited for hidden layers.
- *ReLU (Rectified Linear Unit):* Defined as:

  $f(x) = \max(0, x)$

  ReLU is computationally efficient and helps reduce the vanishing gradient problem.

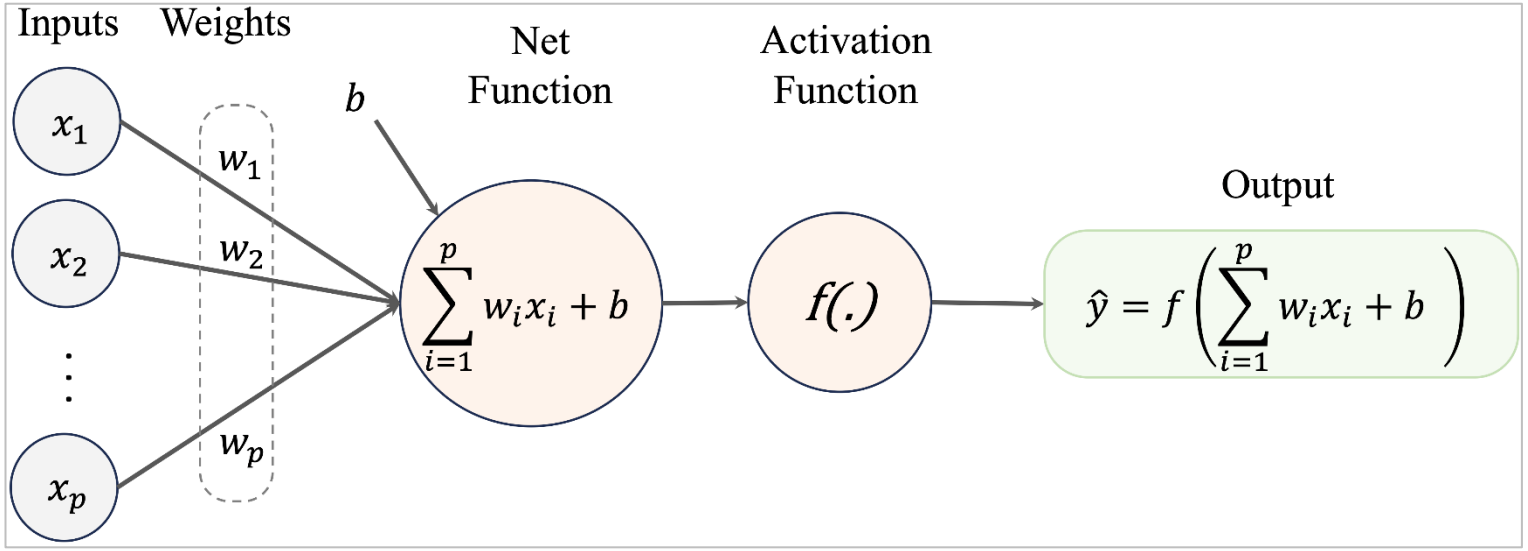
Activation Functions in Neural Networks



UvA

How does a Neural Network learn?

Each node in the network is a neuron having their weights. These weights must be "*learned*" by the model to give the results closest to the training data. This happens through processes called **Back-propagation** and **Gradient Descent**.



Once, we have "trained" the model, ie learned the optimum weights for all neurons, the input data is sent into the Neural Network, each neuron multiplying its inputs and weights, applying the activation on the sum and sending it to the next layer until we have the final result.

UvA

Advantages and Disadvantages of Neural Networks

Advantages:

- Suitable for highly complex settings
- Can handle unstructured data
- Parallel processing
- Can be used for generative purposes

Disadvantages:

- "Black box" nature
- Proneness to overfitting
- Expensive and time-consuming to train
- Sensitive to noise

Jupyter Notebook

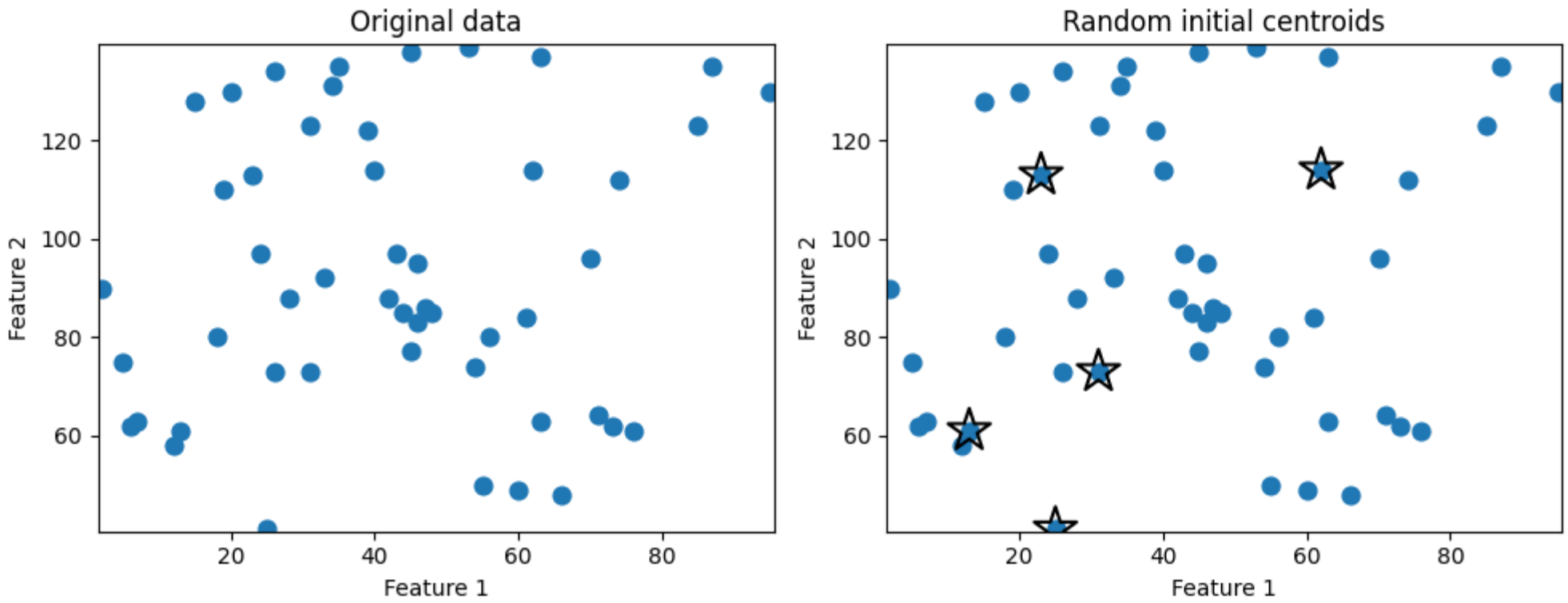
k-Means Clustering

- Unsupervised machine learning technique
- Objective: create *k* clusters
- No labels required
- Greedy algorithm

Steps:

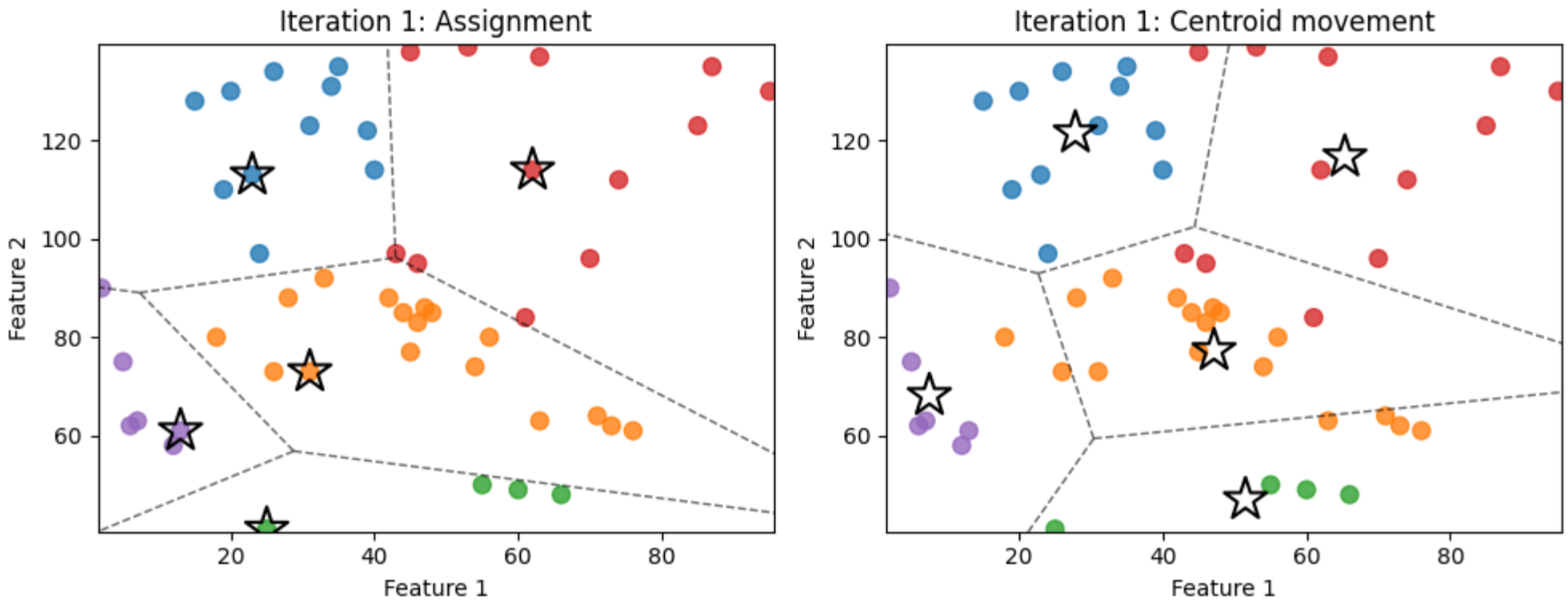
1. Determine number of clusters (k) to be generated
2. Select *k* random starting points as initial centroids
3. Select observations closest to each centroid and assign them to a cluster
4. Compute new centroids (center of each cluster)
5. Repeat steps 3 and 4 until centroids stop moving

Initial Centroids



Five observations are randomly chosen as initial centroids.

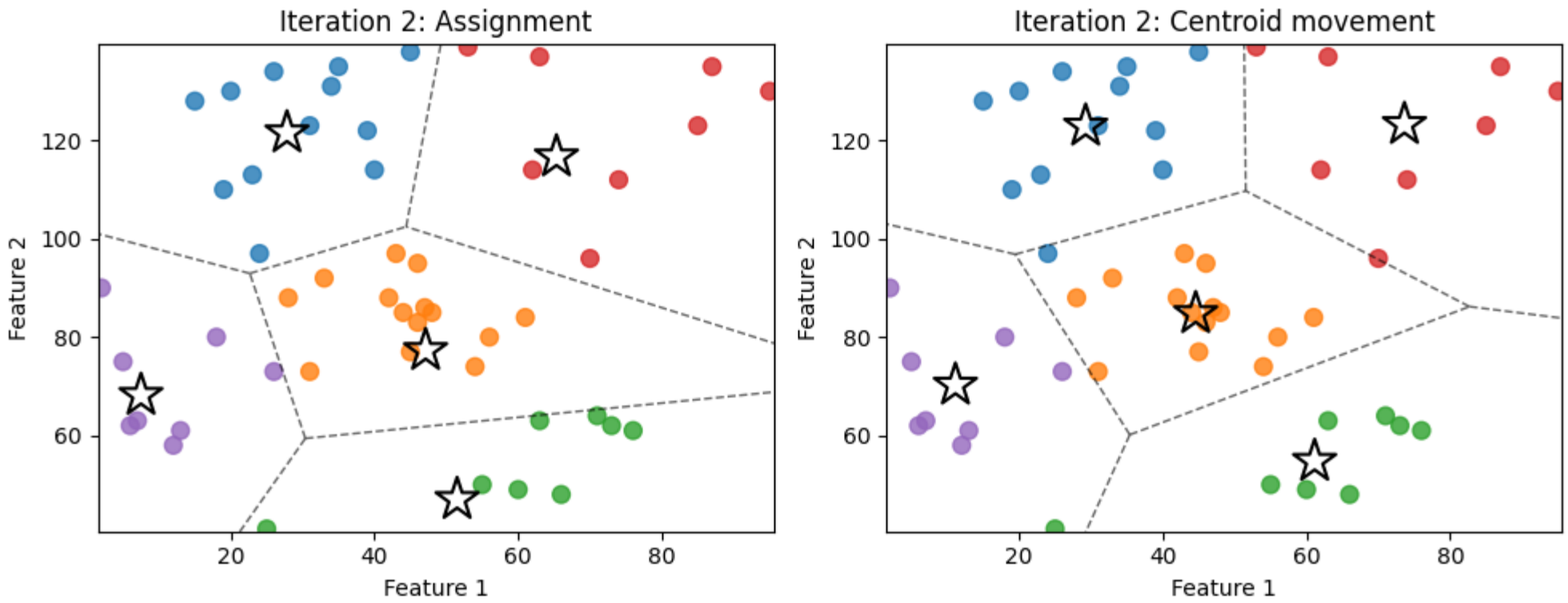
Iteration 1: Assignment and Centroid Movement



1: Each observation is assigned to its nearest centroid, partitioning the feature space into Voronoi regions defined by proximity to the current centroids.
2: Centroids are updated by moving them to the mean position of the observations assigned to each cluster.

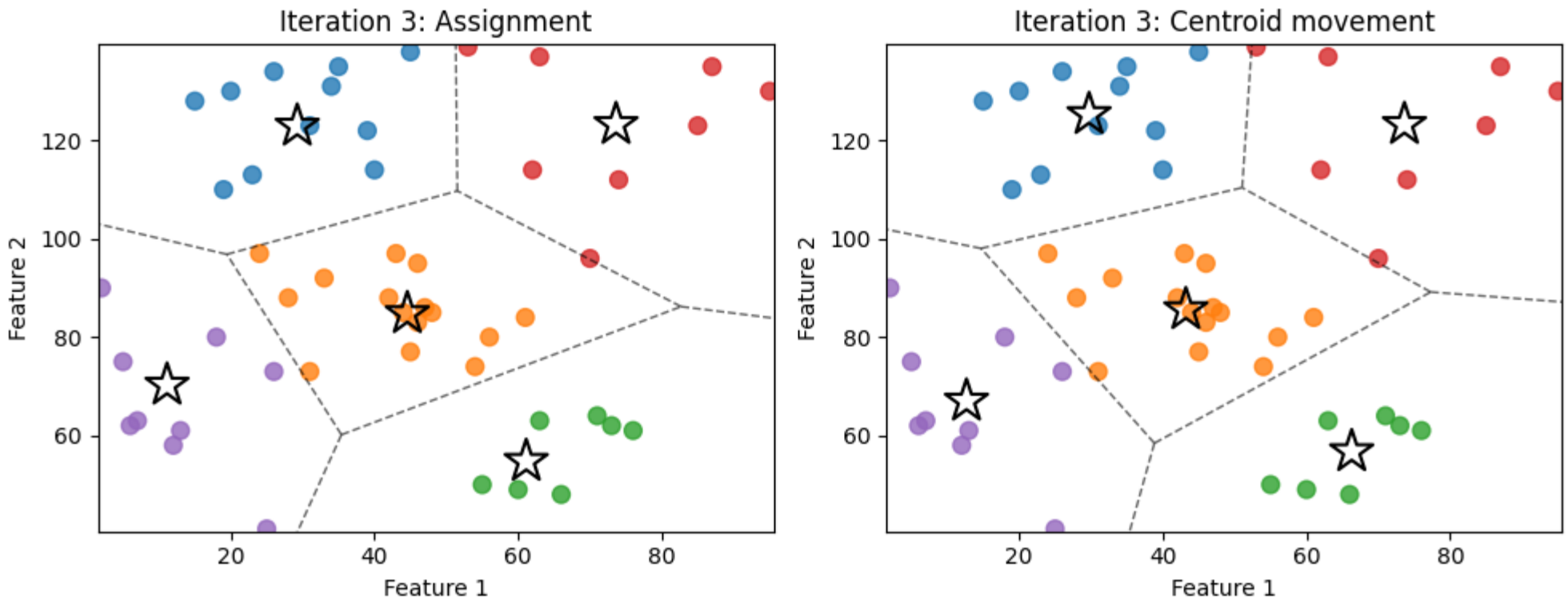
UvA

Iteration 2: Assignment and Centroid Movement



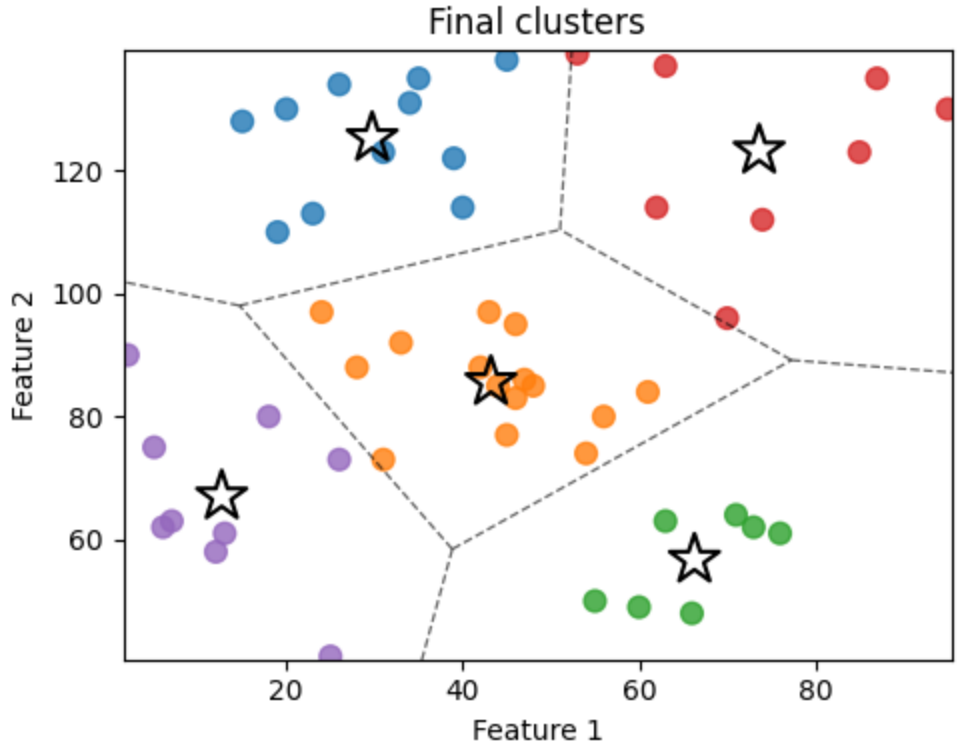
UvA

Iteration 3: Assignment and Centroid Movement



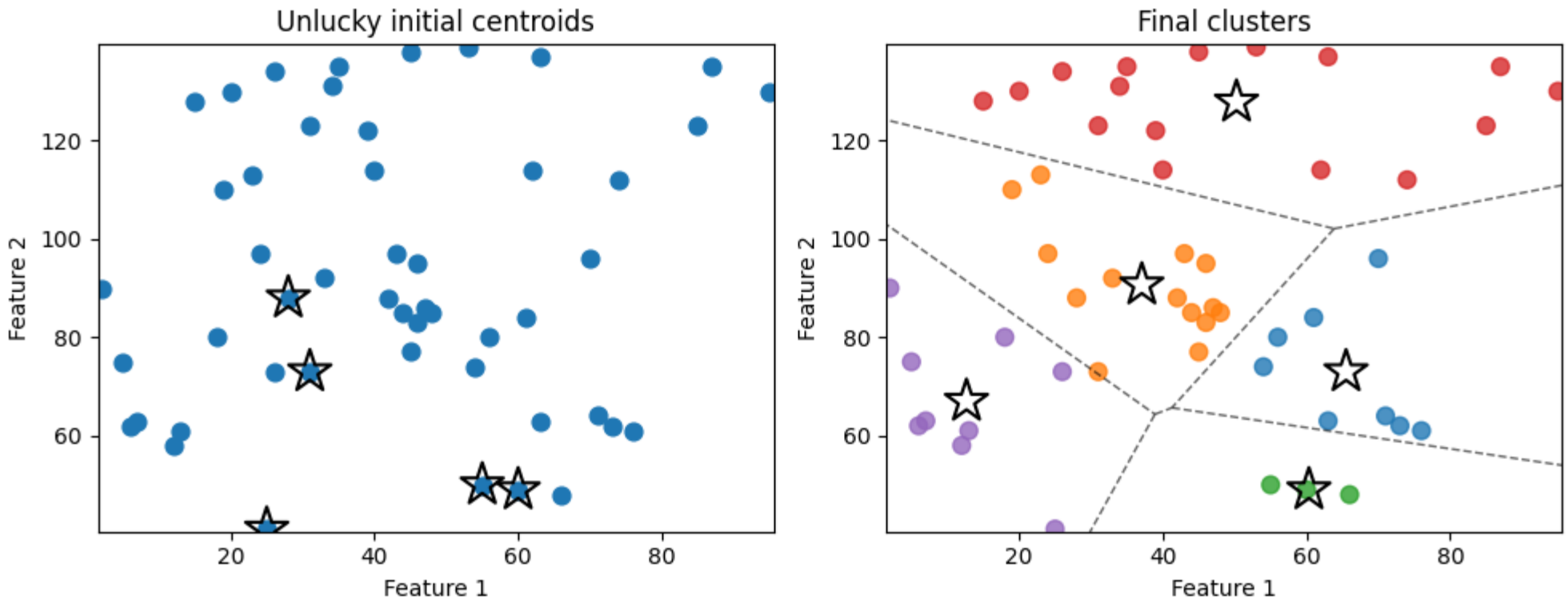
UvA

k-Means Clustering: Final Clusters



UvA

k-Means Clustering: Impact of Unlucky Centroids



UvA

k-Means Illustration: NYC Taxi Zones

- 2025 Yellow Cab Trip Data
  Aggregated at the taxi-zone level

Advantages and Disadvantages of k-Means

Advantages:

- Easy to understand and implement
- Fast algorithm: scales well with large datasets

Disadvantages:

- Depending on the setting, determining the right number of clusters may be subjective
- Greedy algorithm: results dependent on initial centroids
- Size based measures are sensitive to relative scales of features: may need to normalize data (scale: 0-1)

Bonus Challenge: Fraud Recognition

- Image recognition? Digits?
- Fraud?
  - Can extend to imblearn / class weighting

References and Sources

- Chapman, P., Clinton, J., Kerber, R., Khabaza, T., Reinartz, T., Shearer, C., & Wirth, R. (2000). CRISP-DM 1.0: Step-by-step data mining guide. *SPSS inc*, *9*(13), 1-73.

UvA